

Certified Joint Pricing Errors in Rough Heston

Finite History, Actual Outputs, and Financial Decisions

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Abstract

We certify errors of actual rough Heston pricing outputs while retaining the complex Fourier perturbation shared by different strikes. A finite-history theorem propagates a continuous fractional Riccati residual to the derivative-based pricing exponent with its initial curve term and physical scaling intact. It distinguishes a curve-weighted envelope valid under nonexpansion from a sharper positive-dissipation resolvent envelope. Combining certified transform disks with finite omitted frequencies, analytic-strip quadrature, the true infinite tail and arithmetic gives complete price, portfolio and finite-candidate objective enclosures. A matched quarter-year spread comparison changes a quarter-point decision: the joint bound is 0.233318843 index points, whereas the marginal bound is 0.252393939. A fixed five-point nearby roughness grid and direct-reference, corrected-output and modified-Adams controls test the method under unchanged original quotes. The certification budget is decomposed to distinguish history weighting, shared geometry, reference uncertainty and changes of centre. Exact algebraic covers separately establish the specified rational construction on a narrow continuous parameter domain; they are not prerequisites for the independent reference error proof. Integrated appendices retain regularity, residual generation and classical common-state extensions, with explicit verification obligations and incomplete-application boundaries. The guarantee concerns deterministic model-price error, rather than market fit or observed solver error.

Keywords: rough Heston; continuous residual; finite history; actual output; joint pricing error; deterministic certification.

1. Research question and contribution

Can a complete, verifiable bound on a numerical pricing output retain enough shared structure to change a financial certification decision? At one model parameter and maturity, every strike uses the same Fourier transform values. Independently widening each price loses the fact that the perturbation at each frequency is one common complex number. We construct the price-error set from those common variables, rather than infer dependence from observed covariance.

Our object is explicit. Let c^* denote mathematical model prices, c^{fast} the stored production output, and $\bar{c}$ an independent reference centre. The complete inclusion has the form

$$c^* - c^{\text{fast}} = d + \text{Re}(A\delta) + r, \quad d = \bar{c} - c^{\text{fast}}. \quad (1.1)$$

Each component of δ is a rigorously enclosed common Fourier error. Finite omitted frequencies remain in δ ; the remainder r retains strip quadrature, the true infinite tail and arithmetic, without charging those finite nodes twice. A reference refinement changes d and its certified uncertainty together. Neither the reference price nor the fast output is identified with the exact model price.

1.1. Three contributions, with their dependencies

The principal analytical contribution is the connection from a dissipative complex Riccati residual to the Heston exponent through a finite-history kernel. Theorem 3.1 states both the nonexpansive curve bound and the stricter positive-dissipation envelope. The initial curve term and the physical factor ν^{-1} remain explicit. Existing Caputo convexity and positive-resolvent results supply the comparison mechanism; the model-specific pricing-functional composition, certified history weights and complete output budget are the contribution developed here.

The second contribution is a complete implementation-level financial certificate. Shared transform disks are translated to the actual output and evaluated in portfolio and finite-objective directions. The quarter-year matched comparison isolates a certification decision changed by the joint structure. The unified ablation distinguishes propagation, time-local envelopes, omitted-frequency certification and recentering. Support functions, convexity and sorting a fully known unit-cost menu are established tools, rather than separate novelty claims.

The third contribution is a checkable structural domain for the specified Gatheral--Radoicic rational construction, together with an independently generated reference certificate. The all-frequency result has explicit restrictions on correlation, fractional order and mean reversion. The numerical references do not derive their residual certificates from the algebraic sign proof. A nearby-parameter protocol and output controls probe the method's transfer and its appropriate workload.

Proof chain	Inputs and conclusion	Role in the price certificate
Rational construction	Endpoint matching, raw determinant and numerator signs; well-defined Pad�� output and left-half-plane structure	Defines and checks the specified production formula on its stated domain
Independent reference	Fixed field, full continuous residual, dissipation, finite-history exponent and complete Fourier error	Certifies the reference transform and model-price inclusion
Output translation	Stored fast values and independently enclosed reference centre	Converts the reference inclusion into error of the actual output

The chains share the mathematical model. A successful Pad   sign certificate does not by itself certify its pricing error. Conversely, the independent-reference residual route does not require the Pad   trajectory to serve as its reference field.

1.2. Related work and the precise evidence distinction

Gatheral--Radoicic [GR2019, GR2023v1] construct two-end rational approximations. Jeng--Kilicman [JK2020, JK2021] analyse global Pad   approximations and public SPX data. The formula and endpoint-matching method used here are theirs. Our structural statement concerns the raw matching system throughout its explicitly certified domain.

Abi Jaber--El Euch [AbiJaberElEuch2018v1] supply the Volterra model and affine transform. Li--Liu [LiLiu2018] establish Caputo convexity through regularization; Kopteva [Kopteva2021v2] propagates a time-dependent residual with a positive fractional inverse. Simon [SimonCM2015] supplies Mittag--Leffler positivity, and Trefethen--Weideman [TrefethenWeideman2014] supplies strip trapezoidal analysis. We state the exact regularity bridge consumed here and attribute these mechanisms.

Ben Hammouda et al. [BenHammouda2026v1, Section 3.2] develop error-controlled multilevel rough-Heston quadrature. In their practical tolerance procedure, unavailable constants are replaced by numerical error indicators, and tolerance attainment is assessed numerically rather than certified a posteriori. Their useful algorithmic error-control objective differs from our complete deterministic enclosure of a stored output, including tails and arithmetic. This distinction is specific to the cited version and procedure; it is not a claim that all other rough-Heston methods lack rigorous analysis.

Boyarchenko--de Innocentis--Levendorskii [BL2025v1] address reliable pricing, numerical bias and incorrect calibration. Their modified Adams method is a relevant numerical comparison. Their Conformal Bootstrap is explicitly an ad-hoc principle, and is not treated here as a deterministic interval guarantee. Hager--Kreher [HK2026v1] develop analytic expansions in the Hurst parameter and local convergence results; those results address a different approximation object. Bayer--Breneis [BBWeak2023v1, BBSimulation2023v1] control Markovian kernel approximations and investigate low-dimensional simulation. Comparisons therefore state the model, output and guarantee being compared.

This bounded literature comparison supports the particular contribution claimed above. It does not establish priority over every unpublished or contemporaneous method. The classical Heston original-chain analysis is preserved as a separate

extension, with its own probability objects and incomplete annual monetary certificate; it is not a main contribution of this rough-Heston paper.

2. Model, units and fixed configurations

Our conventions for the fractional integral and Caputo derivative are

$$I^\alpha f(t) = \frac{1}{\Gamma(\alpha)} \int_0^t (t-s)^{\alpha-1} f(s) ds, \quad D_C^\alpha v = I^{1-\alpha} v', \quad v \in AC. \quad (2.1)$$

Under the regularity of the trajectories used below, $I^\alpha f \in AC$ with zero initial value, so $D_C^\alpha I^\alpha f = f$. The Caputo operator and all time scalings retain their fractional definitions. Set $a = u - i/2$. The physical Riccati state h satisfies

$$D_{C,t}^\alpha h = -\frac{a^2 + ia}{2} + (i\rho\nu a - \lambda_R)h + \frac{\nu^2}{2}h^2, \quad h(0) = 0. \quad (2.2)$$

Define

$$\begin{aligned} x &= \nu^{1/\alpha} t, \quad y = x^\alpha = \nu t^\alpha, \quad H(x) = \nu h(t), \quad Z(t) = H(\nu^{1/\alpha} t), \quad \kappa = \lambda_R/\nu, \\ b &= (u^2 + 1/4)/2, \quad s_0 = \kappa - \rho/2, \quad d = -s_0 + i\rho u, \quad F(z) = -b + dz + z^2/2. \end{aligned} \quad (2.3)$$

Then $D_{C,x}^\alpha H = F(H)$ and $D_{C,t}^\alpha Z = \nu F(Z)$. Dividing the normalised state error by ν gives the physical h error. Likewise, dividing the physical residual $r_t = D_t^\alpha \widehat{Z} - \nu F(\widehat{Z})$ by ν gives the residual bound for the normalised equation.

For pricing and the numerical example, we fix $\kappa = 0$ and the entire forward variance curve

$$\xi_*(t) = \theta + (V_0 - \theta)E_{\alpha_0}(-\lambda_\xi t^{\alpha_0}), \quad (\alpha_0, V_0, \theta, \lambda_\xi) = (.5286, .0262, .0721, .5037). \quad (2.4)$$

The curve parameter λ_ξ and Riccati parameter λ_R are defined separately. Curve (2.4) remains fixed when the candidate α changes. The probability model is

$$V_t = \xi_*(t) + \nu \int_0^t K_\alpha(t-s) \sqrt{V_s} dW_s, \quad K_\alpha(t) = t^{\alpha-1}/\Gamma(\alpha), \quad dS_t = S_t \sqrt{V_t} dB_t, \quad d\langle B, W \rangle_t = \rho dt. \quad (2.5)$$

Equation (2.5) is the forward variance representation with zero Riccati mean reversion. Appendix A verifies the probability model and affine transform associated with this curve using Theorems 2.1 and 2.3 and Example 2.2 of Abi Jaber and El Euch.

Let $M_T = S_T/F_T$ be the normalised positive martingale and write $\phi_T(a) = \mathbb{E}M_T^{ia}$. The exact characteristic exponent is

$$L_T(u) = \int_0^T \xi_*(T-t) F(Z(t, u)) dt, \quad \phi_T(u - i/2) = e^{L_T(u)}. \quad (2.6)$$

With $m = K/F_T, k = \log m, c = C/(DF_T)$, the model European call price is

$$c = 1 - \frac{\sqrt{m}}{\pi} \int_0^\infty \operatorname{Re} \frac{e^{-iuk} \phi_T(u - i/2)}{u^2 + 1/4} du. \quad (2.7)$$

The model specification, decimal inputs, and normalisation jointly define the prices considered below.

All experiments use model parameter $\nu = 0.2897$, $\rho = -0.7445$, Riccati mean reversion zero, Fourier step $1/8$, a finite reference/error grid through 128 (1025 nodes), 100-bit outward dyadic arithmetic and 64 forward-moment series terms. The actual Pad   output uses composite Gauss-Legendre order 8 through 200 (352 nodes) and Jacobi order 256; its Fourier cutoff is distinct from the certificate cutoff. N_t counts source time cells. The number of frequency nodes is written explicitly to avoid confusing it with ν .

ID	T	alpha	N't source cells	nonzero ref nodes	zero finite nodes	subcells	history bins
H0	1/2	13/25, 3/5, 9/10	2048 / 2048 / 1024	513	512	4	0
H1	1/2	13/25, 3/5, 9/10	2048 / 2048 / 1024	513	512	4	1
H2	1/2	13/25	2048	513	512	4	128
H3	1/2	13/25	2048	1025	0	4	low128; high1
H4	1/2	13/25	2048	1025	0	4	128
Q0	1/4	13/25, 3/5, 9/10	1429 / 1352 / 549	513	512	4	1
Q1	1/4	13/25	1429	1025	0	4	1
Q2	1/4	13/25	1429	1025	0	4	128
N1	1/2	13/25, 21/40, 53/100, 27/50, 11/20	1024	513	512	2	1
N2	1/2	13/25, 21/40, 53/100, 27/50, 11/20	2048	513	512	2	1
N2 L	1/2	13/25	2048	513	512	2	128

H0 uses the old uniform-state propagation; H1 uses finite-horizon global propagation on the identical field. H2 retains the original low-frequency 128-bin bank. H3/H4 expand the same reference centre through 128, retaining that low bank and adding the newly certified high bank with global/128-bin propagation. Q0/Q1/Q2 keep full history from zero and use exact quarter-terminal interpolation of the original half-year field: Q0 has 1430/1353/550 time knots, and Q1/Q2 have 1430 knots. Their larger-horizon continuous banks are reused; quarter moments, output and tails are recomputed. The original alpha=.9 source field has 1025 time knots and 641 stored frequencies through 80; only 513 are nonzero reference nodes in H0/H1/Q0. The .52/.60 original fields have 2049 knots and 1025 stored frequencies through 128.

N1/N2 regenerate each five-point field and its continuous residual bank, with two nonstartup subcells (2047/4095 closed cells). N2L is a separately frozen descriptive 128-bin reuse of N2 at alpha=.52; it does not replace the nearby-grid objective

experiment. The old low and high banks each have 8189 closed cells (four nonstartup subcells), for 513 and 512 frequencies respectively. Thus the 0.115215934-point H4 certificate and 0.394999331-point N2L certificate concern different reference centres, node coverage and residual banks. Their difference cannot be attributed solely to 128-bin propagation. BL-core512/1024 are nominal history-stepping controls whose complete guarantees use the explicitly identified N1/N2/N2L reference bank.

The H and N field generators use a uniform auxiliary grid y_j from zero to νT^α , followed by $x_j = y_j^{1/\alpha}$ and $t_j = x_j/\nu^{1/\alpha}$. In exact arithmetic this is the graded rule $t_j = T(j/N_t)^{1/\alpha}$. Generation uses binary64 arithmetic and forces the endpoints to zero and one half; certification uses the saved nodes as exact dyadics, rather than identifying them with the analytical rule. For quarter-year restriction it keeps every original knot strictly below $T_q = 1/4$, appends the exact endpoint, and encloses the linear interpolation of the original bracketing field coefficients in rational interval arithmetic. All preceding history is retained.

3. Finite-history propagation: nonexpansion and dissipation

The scalar Caputo comparison and norm-convexity mechanisms are established tools [LiLiu2018; Kopteva2021v2; SimonCM2015]. Appendix C.1 retains the full V2 absolute-continuity proof, including almost-everywhere passage and the continuous endpoint. Under the hypotheses below the complex divided difference has real part at most $-\nu\sigma$. The model-specific composition with the pricing functional yields the following theorem. Its proof occurs only in Appendix C.2.

For $\beta > 0$, write $g_\beta(t) = t^{\beta-1}/\Gamma(\beta)$. Define $k_\lambda(t) = t^{\alpha-1}E_{\alpha,\alpha}(-\lambda t^\alpha)$, with $k_0 = g_\alpha$. All convolutions below start at time zero.

Theorem 3.1 (finite-history pricing envelope). Let $0 < \alpha < 1$, $\nu > 0$, $T > 0$, and $Z, \widehat{Z} \in AC([0, T]; \mathbb{C})$ have zero initial value. Let

$$D_C^\alpha Z = \nu F(Z), \quad r = D_C^\alpha \widehat{Z} - \nu F(\widehat{Z}), \quad |r| \leq R \in L^\infty(0, T), \quad R \geq 0, \quad (3.1)$$

where $F(z) = -b + dz + z^2/2$, $\Re d = -s_0$, $\Re Z \leq 0$, $\Re \widehat{Z} \leq \epsilon_R$, and $\sigma = s_0 - \epsilon_R/2 \geq 0$. These equations and inequalities hold almost everywhere; the state bounds hold everywhere by continuity. Let $\xi \in AC[0, T]$, $\xi(0) = V_0 \geq 0$, and assume

$$q_\alpha = (I^{1-\alpha}\xi)' = V_0 g_{1-\alpha} + g_{1-\alpha} * \xi' \geq 0 \quad \text{a.e.} \quad (3.2)$$

Define L_T and $\widehat{L}_T$ from $\nu^{-1} \int_0^T \xi(T-t) D_C^\alpha Z(t) dt$ and the same expression with $\widehat{Z}$. With $\lambda = \nu\sigma$,

$$|Z - \widehat{Z}| \leq k_\lambda * R, \quad |L_T - \widehat{L}_T| \leq \nu^{-1}(q_\alpha * k_\lambda * R)(T) \leq \nu^{-1}(\xi * R)(T). \quad (3.3)$$

In particular, $R/\nu \leq \delta_F$ implies

$$|L_T - \widehat{L}_T| \leq \delta_F \int_0^T \xi(s) ds. \quad (3.4)$$

At $\sigma = 0$, set $k_0 = g_\alpha$; the first and second exponent envelopes coincide. Strict positive dissipation is required for an additional reduction, rather than for validity of the curve-weighted bound. The initial term $V_0 g_{1-\alpha}$ and the factor ν^{-1} cannot be dropped. Curve admissibility for this analytical theorem is distinct from stochastic-model admissibility, verified in Appendix A.

For a full-history residual envelope $R \leq R_j$ on each closed cell $[a_j, b_j]$, the computable cell-weight version is

$$|L_T - \widehat{L}_T| \leq \nu^{-1} \sum_j R_j \int_{a_j}^{b_j} (q_\alpha * k_\lambda)(T - s) ds \leq \nu^{-1} \sum_j R_j \int_{a_j}^{b_j} \xi(T - s) ds. \quad (3.5)$$

Every weight is a rigorous integral over the history measured from time zero. An integration bin uses the maximum over every intersecting closed residual cell, including boundary cells. The original experiments use curve weights; Section 7 identifies the new matched dissipative-kernel experiment explicitly. Analytical decreasing-curve and constant-curve variants, and the complementary global-state bound, are proved in Appendix C.3-C.4.

Strict computation of the dissipative pricing kernel

The positive kernel in Theorem 3.1 can be computed rather than replaced by its curve upper bound. For $\sigma \geq 0$, set $\lambda = \nu\sigma$ and

$$K_\lambda = q_\alpha * k_\lambda, \quad \eta_{\text{res}} = \nu^{-1}(K_\lambda * R)(T), \quad 0 \leq K_\lambda = \xi - \lambda \xi * k_\lambda \leq \xi. \quad (3.6)$$

The complete residual history, zero state initial values, curve initial term and physical factor ν^{-1} remain those of Theorem 3.1. The scalar inverse and its positivity are established tools [Kopteva2021v2, SimonCM2015]. The additional implementation here rigorously encloses the model's pricing-kernel weights and carries them through the same actual-output certificate.

Corollary (nonexpansive finite-history propagation). Theorem 3.1 remains valid with $\sigma \geq 0$. At $\sigma = 0$,

$$k_0 = g_\alpha, \quad K_0 = \xi, \quad |L_T - \widehat{L}_T| \leq \nu^{-1}(\xi * R)(T). \quad (3.7)$$

No reciprocal damping or infinite-horizon state radius is needed. The curve condition is $q_\alpha \geq 0$, not $\xi' \geq 0$; the stochastic-model hypotheses remain separate. The existing AC regularity bridge and full comparison proof are retained in Appendix C.

For the fixed curve in (2.4), write $A_0 = \alpha_0$ and λ_ξ for its own curve parameter. On the computational domain $\lambda T^\alpha < 1$, define

$$W_\lambda(t) = \int_0^t K_\lambda(s) ds = \sum_{n=0}^{\infty} (-\lambda)^n t^{1+n\alpha} \left[\frac{\theta}{\Gamma(2+n\alpha)} + (V_0 - \theta) E_{A_0, 2+n\alpha}(-\lambda_\xi t^{A_0}) \right]. \quad (3.8)$$

A cell $[a_j, b_j]$ has weight $W_\lambda(T - a_j) - W_\lambda(T - b_j)$. Appendix C provides the absolute series tails and interval-difference proof. The implementation uses twelve outer terms, sixty-four inner curve terms and 100-bit outward dyadic primitives. It is an enclosure of

the full Mittag-Leffler kernel functional, with explicit truncation and rounding remainders; ordinary floating-point Mittag-Leffler values do not enter a guarantee.

4. Continuous certificates and complete Fourier prices

The reference is a fixed $\widehat{Z} = I^\alpha \bar{G}$, with recorded coefficients interpreted as exact dyadics; this continuous reference is not identified with the nominal nodal solver states. A startup power expansion and closed-cell derivative bounds certify the full physical residual. Appendix D.1 gives the complete construction, arithmetic and history integrals. Four-subcell historical banks and two-subcell nearby banks are distinct valid covers, identified in the configuration table.

The derivative-based reference exponent is $\widehat{L}_T = \nu^{-1} \int_0^T \xi(T-t) \bar{G}(t) dt$. Once η_n encloses its exponent error, the transform disk follows by the exponential difference and the true half-shift modulus bound. A substitution-based exponent would require an additional residual-conversion term.

$$|\phi_n - \widehat{\phi}_n| \leq \min\{1, |\widehat{\phi}_n|\}(e^{\eta_n} - 1). \quad (4.1)$$

Theorem 4.1. Suppose $M_T > 0, \mathbb{E}M_T = 1$. For any $0 < a_* < 1/2, h_* > 0$, let $g(z) = e^{-ikz} \phi_T(z - i/2)/(z^2 + 1/4)$. Replacing the integral in (2.7) by the infinite trapezoidal sum incurs a price error of at most

$$\epsilon_{\text{grid}} = \frac{\sqrt{m} e^{a_*|k|}}{(1/2 - a_*)(e^{2\pi a_*/h_*} - 1)}. \quad (4.2)$$

The strip proof and the exact-solution high-frequency comparison are in Appendix D.2. A finite omitted reference value is explicitly charged, independently of the true infinite tail.

Suppose the node values $\widehat{\phi}_n$ have exact-error bounds ε_n . The finite price sum

$$\widehat{c}_{N_u} = 1 - \frac{h_* \sqrt{m}}{\pi} \left(2 \operatorname{Re} \widehat{\phi}_0 + \sum_{n=1}^{N_u} \frac{\operatorname{Re}(e^{-inh_* k} \widehat{\phi}_n)}{(nh_*)^2 + 1/4} \right) \quad (4.3)$$

satisfies

$$|c - \widehat{c}_{N_u}| \leq \epsilon_{\text{grid}} + \epsilon_{\text{tail}} + \frac{h_* \sqrt{m}}{\pi} \left(2\varepsilon_0 + \sum_{n=1}^{N_u} \frac{\varepsilon_n}{(nh_*)^2 + 1/4} \right) + \epsilon_{\text{arithmetic}}. \quad (4.4)$$

In these equations N_u is the Fourier sum cutoff, whereas N_t denotes time-grid cells. The origin coefficient is the trapezoidal half-weight multiplied by $1/(1/4)$. The exact arithmetic remainder encloses phase, square root, exponential and summation operations.

5. Shared errors and financial decisions

5.1. Joint inclusion at the frozen output

Fix one model parameter, maturity, contour, and reference grid. Let $c^* \in \mathbb{R}^p$ be the continuous-model price vector, $\bar{c}$ the exact finite reference sum, and c^{fast} the frozen rational output. For each included frequency let $\bar{\phi}_n$ be the reference transform and put $z_n = \phi(u_n - i/2) - \bar{\phi}_n$. Appendix D.1 supplies bounds $|z_n| \leq \epsilon_n$; at a deliberately omitted finite node $\bar{\phi}_n = 0$, the true-transform envelope supplies the radius. The same z_n enters every strike at this parameter and maturity.

For the Lewis rule in Section 4, set $m_i = K_i/F$, $k_i = \log m_i$, and

$$a_{in} = -\frac{h\sqrt{m_i}}{\pi} \frac{e^{-iu_n k_i}}{u_n^2 + 1/4} \quad (n > 0), \quad a_{i0} = -\frac{2h\sqrt{m_i}}{\pi}. \quad (5.1)$$

The half weight at zero is already included in a_{i0} . The complete error is

$$c^* - \bar{c} = \Re \sum_{n=0}^N a_{\cdot n} z_n + R, \quad R \in \mathcal{R}, \quad (5.2)$$

where $\mathcal{R}$ includes the true infinite-grid tail and analytic-strip discretisation error. A finite reference sum enclosed by interval arithmetic contributes its centre uncertainty when a numerical representative of $\bar{c}$ is used. Omitted finite nodes remain in (5.2); they are not also charged as an infinite tail. No stochastic covariance is used in this representation.

Theorem 5.1 (complete shared-node inclusion). Suppose (5.2) holds, all radii are nonnegative, and $\mathcal{R}$ is a nonempty compact convex outer set. Define

$$\mathcal{E}_F = \left\{ \Re \sum_n a_{\cdot n} z_n : |z_n| \leq \epsilon_n \right\} + \mathcal{R}, \quad d = \bar{c} - c^{\text{fast}}. \quad (5.3)$$

Then $c^* - c^{\text{fast}} \in d + \mathcal{E}_F$, and, for real w ,

$$h_{d+\mathcal{E}_F}(w) = w^\top d + \sum_n \epsilon_n \left| \sum_i w_i a_{in} \right| + h_{\mathcal{R}}(w). \quad (5.4)$$

If the centre shift is supplied as a box $d \in [d^-, d^+]$, replace the first term by the support of that box. All numerical evaluations of (5.4) must be outward enclosures.

Proof. Substituting the actual transform errors in (5.2) gives inclusion. The product of complex disks is compact and convex and its real-linear image has those properties. A disk $|z| \leq \epsilon$ has real-linear support $\epsilon|b|$ for the functional $\Re(bz)$: Cauchy-Schwarz gives the upper bound, attained at $z = \epsilon \bar{b}/|b|$ when $b \neq 0$; when $b = 0$, every feasible point attains it. Independent disks and Minkowski addition give (5.4). Translation gives the signed centre term. A box enclosing d provides a further valid Minkowski outer enclosure even if its coordinates are dependent. $\square$

If $\mathcal{R} = \prod_i [-\rho_i, \rho_i]$, the smallest coordinate box of this same set has support

$$h_{\text{rect}(\mathcal{E}_F)}(w) = \sum_i |w_i| \left(\sum_n \epsilon_n |a_{in}| + \rho_i \right). \quad (5.5)$$

Consequently its excess over (5.4), before translation, is

$$G_F(w) = \sum_n \epsilon_n \left(\sum_i |w_i a_{in}| - \left| \sum_i w_i a_{in} \right| \right) \geq 0. \quad (5.6)$$

It is strictly positive exactly when one positive-radius node has nonzero coefficients $w_i a_{in}$ that do not all lie on a common nonnegative complex ray. This follows from the equality case of the complex triangle inequality, node by node. Translation does not change widths. This criterion compares the constructed set with its own coordinate box, and does not assert strict improvement over every independently available signed price enclosure.

For any separately established signed model-price box $\mathcal{I}$, intersect it with $\bar{c} + \mathcal{E}_F$. The actual model vector belongs to the intersection. Without solving the intersection support problem, the minimum of the two valid directional upper bounds remains valid; likewise take the maximum of the lower bounds. Thus a computation can retain the best signed marginal information while using node cancellation where it improves the complete bound. Nonemptiness of the mathematical intersection follows from inclusion, rather than from empirical agreement between numerical approximations.

Proposition 5.2 (nonzero omitted-node reference at an unchanged fast output). Fix the actual fast output, including its zero contribution at omitted finite nodes. For the same finite pricing map, select a possibly nonzero reference ψ_n at those nodes, and prove $|\phi_n - \psi_n| \leq \rho_n$. Let $\bar{c}' = c_0 + \Re(A\psi)$, $d' = \bar{c}' - c^{\text{fast}}$, and let $\mathcal{R}$ include the complete strip, true infinite tail, and newly enclosed reference arithmetic. Then

$$c^* - c^{\text{fast}} \in d' + \{\Re(Az) : |z_n| \leq \rho_n\} \oplus \mathcal{R}. \quad (5.7)$$

For a symmetric remainder and direction w , the complete absolute budget uses

$$DF \left(|w^\top d'| + \sum_n \rho_n \left| \sum_i w_i a_{in} \right| + h_{\mathcal{R}}(w) \right) \leq \tau. \quad (5.8)$$

For a nonsymmetric remainder, both support directions must be bounded. A changed reference centre does not change the fixed fast output, but it does change d' and requires new centre arithmetic.

Proof. Subtract and add the new finite reference in the exact finite pricing map; its transform error is $z = \phi - \psi$. The complete remainder contains the other discrepancies. Directional support of the shared complex disks gives (5.8). ■

An old zero-centred certificate $|\phi_n| \leq \varepsilon_n^0$ and a new certificate centred at ψ_n cannot have their radii directly minimized. At the new centre, the old certificate only gives $|\phi_n - \psi_n| \leq \varepsilon_n^0 + |\psi_n|$. A safe radius is therefore $\min\{\rho_n, \varepsilon_n^0 + |\psi_n|\}$, or one can retain the two transform disks' intersection. A new disk is nested in the old disk only if $|\psi_n| + \rho_n \leq \varepsilon_n^0$. Otherwise, independently valid complete price intervals may still be intersected, but same-centre monotonicity is not automatic. For example, $\phi = 0$ lies in

$D(0, .1)$ and $D(1, 2)$, but not in $D(1, \min(.1, 2))$. Marginal and joint comparisons must use the same updated centre, node radii, and complete remainder. Any additional reference evaluation and certification is part of the reported cost.

5.2. Directional and finite-objective certificates

For $w = e_i - e_j$, the node coefficient is

$$|a_{in} - a_{jn}| = \frac{h}{\pi(u_n^2 + 1/4)} \left| \sqrt{m_i} e^{-iu_n k_i} - \sqrt{m_j} e^{-iu_n k_j} \right| \quad (n > 0). \quad (5.9)$$

It is the modulus of one combined coefficient, rather than the sum of two moduli. A useful analytic bound, valid with $b_i = \sqrt{m_i}$, is

$$|b_i e^{-iuk_i} - b_j e^{-iuk_j}| \leq |b_i - b_j| + \min(b_i, b_j) \min\{2, |u| |k_i - k_j|\}. \quad (5.10)$$

To prove it, separate the difference in amplitudes from the phase difference using the smaller amplitude, then apply $|e^{ix} - e^{iy}| \leq \min(2, |x - y|)$. At zero, the spread coefficient is $2h|b_i - b_j|/\pi$. Low-frequency error therefore cancels for nearby strikes even when the individual node radii are large. High-frequency and tail contributions are still retained.

The implementation below uses rational outward bounds for logarithms, trigonometric functions, square roots, and the circle constant. It encloses the exact coefficient difference before multiplying by the saved radius. Reference-sum and frozen-output intervals supply the signed centre. Taking interval midpoints for display never substitutes for the outward endpoint arithmetic used in a decision.

If the target quote m^* is enclosed around a stored centre $\bar{m}$ with $m^* - \bar{m} \in \mathcal{M}$, use $r = c^{\text{fast}} - \bar{m}$ and replace the output-error set by $\mathcal{E} - \mathcal{M}$. This includes quote-conversion arithmetic with the correct sign. The following formulas otherwise take m to be exact.

Let $J(e) = (r + e)^\top W(r + e)/(2p)$, where $r = c^{\text{fast}} - m$, $W \succeq 0$, and the complete output error belongs to a compact convex set $\mathcal{E}$. For any trial vector e_0 , put $g_0 = W(r + e_0)/p$. Convexity gives the certified lower bound

$$\inf_{e \in \mathcal{E}} J(e) \geq J(e_0) - g_0^\top e_0 - h_{\mathcal{E}}(-g_0). \quad (5.11)$$

This follows by taking the infimum of the supporting affine function $J(e_0) + g_0^\top (e - e_0)$. Feasibility of e_0 is unnecessary for validity; it affects sharpness. A numerical primal minimizer becomes a lower-bound certificate only after its support or dual obligation has also been bounded correctly.

If $M^2 \geq \sup_{e \in \mathcal{E}} e^\top W e$, expansion yields

$$\sup_{e \in \mathcal{E}} J(e) \leq J(0) + \frac{h_{\mathcal{E}}(Wr)}{p} + \frac{M^2}{2p}. \quad (5.12)$$

A safe choice is any valid coordinate box $|e_i| \leq b_i$: with W represented exactly, $M^2 = \sum_{ij} |W_{ij}| b_i b_j$ is sufficient. Tighter validated norm bounds may replace it. Maximization of this convex quadratic is not certified by a local stationary point. A supporting function, norm bound, interval subdivision, or valid relaxation must establish the upper direction.

For a finite candidate collection, form valid objective intervals by intersecting (5.11)–(5.12), when used, with the original interval-square bounds in Theorem 5.3. A unique candidate is certified only if its upper endpoint is smaller than every competing lower endpoint. If this condition is absent, return the nonempty set of candidates whose lower endpoint is at most the least upper endpoint. It contains every exact minimizer. Different candidates have different transform errors; (5.4) does not identify their error variables.

The support formula, the equality case of the triangle inequality, and the convexity bounds are established tools. Their role here is to join the model-specific continuous residual certificate to the actual multi-strike pricing output, preserving every required budget and its centre. The numerical comparison in Section 7 is the evidence for the resulting financial improvement.

For Theorem 5.3 we specialize the preceding general objective to $p = 12, W = I_{12}$ and sum exactly twelve normalized call prices, so $J = \frac{1}{24} \sum_{i=1}^{12} (c_i - M_i)^2$. Quote-conversion half-widths concern the input midpoint arithmetic and are distinct from the resulting objective half-width.

Theorem 5.3 (finite-objective intervals and selection stability). For each $\alpha_j \in \Theta_{\text{finite}}$, suppose all model prices have certified enclosures

$$c_i(\alpha_j) \in [p_{ij}^-, p_{ij}^+], \quad B_i \in [b_i^-, b_i^+], \quad A_i \in [a_i^-, a_i^+], \quad M_i \in [m_i^-, m_i^+]. \quad (5.13)$$

Assume $b_i^- \leq b_i^+ \leq a_i^- \leq a_i^+$. Define

$$\ell(x, y) = \begin{cases} 0, & x \leq 0 \leq y, \\ \min(x^2, y^2), & \text{otherwise,} \end{cases} \quad v(x, y) = \max(x^2, y^2).$$

$$L_j = \frac{1}{24} \sum_i \ell(p_{ij}^- - m_i^+, p_{ij}^+ - m_i^-), \quad U_j = \frac{1}{24} \sum_i v(p_{ij}^- - m_i^+, p_{ij}^+ - m_i^-). \quad (5.14)$$

Then $J(\alpha_j) \in [L_j, U_j]$. Writing $L_* = \min_j L_j, U_* = \min_j U_j$, the exact finite-set optimum satisfies

$$J_* = \min_{\Theta_{\text{finite}}} J \in [L_*, U_*]. \quad (5.15)$$

For $\varepsilon \geq 0$, every exact ε -near-optimal candidate belongs to

$$\{\alpha_j : L_j \leq U_* + \varepsilon\}; \quad (5.16)$$

The condition $U_j - L_* \leq \varepsilon$ is sufficient for that candidate to be near-optimal for the exact objective. If

$$g := \min_{j \neq j_0} L_j - U_{j_0} > 0, \quad (5.17)$$

then α_{j_0} is the unique minimiser of the model objective on the finite set. If

$$\min_{\alpha_j \geq .6} L_j > U_* + \varepsilon, \quad (5.18)$$

then every exact near-optimal candidate in the finite set satisfies $\alpha_j < .6$.

Quote consistency is checked separately. If any row satisfies $p_{ij}^+ < b_i^-$ or $p_{ij}^- > a_i^+$, the candidate cannot lie in all quote bands simultaneously. Conversely, if every row satisfies $p_{ij}^- \geq b_i^+$ and $p_{ij}^+ \leq a_i^-$, then quote consistency is certified. Overlapping intervals without this inner-band condition retain possible consistency; overlap alone does not certify it.

The interval-square and finite-selection proof is in Appendix D.4. A continuous parameter optimum, market identification and dependence across candidate errors are outside this statement.

6. Structure of the specified rational output

The Gatheral-Radoicic third-order construction matches three startup and three large-time coefficients. Write $\hat{H}(y) = P(y)/Q(y)$, $Q = 1 + q_1y + q_2y^2 + q_3y^3$, with $y = \nu t^\alpha$. Appendix B.1 defines the raw matching matrix, determinant and numerator polynomials, proves their exact identities and gives the full sign proof. No numerical division precedes the determinant lower bound.

Theorem 6.1 (full-frequency structure). For

$$\alpha \in [13/25, 3/5], \quad \rho = -1489/2000, \quad \kappa = 0, \quad u \in \mathbb{R}, \quad \nu > 0, \quad (6.1)$$

the established matching system (B.4) is nonsingular. Its normalised denominator satisfies

$$\operatorname{Re} q_j > 0 \ (j = 1, 2, 3), \quad \operatorname{Re} Q(y) \geq 1, \quad |Q(y)| \geq 1 \ (y \geq 0). \quad (6.2)$$

Moreover, for $y > 0$, $\operatorname{Re} \hat{H}(y) < 0$. The result holds at every positive time and every finite real frequency. Its domain is the stated parameter set; $\alpha > .6$, a continuous ρ domain, and nonzero κ are outside the scope of this theorem.

Theorem 6.2 (certified correlation persistence). The conclusions of Theorem 6.1 hold for

$$\alpha \in [13/25, 3/5], \quad \rho \in [-744501/10^6, -744499/10^6], \quad \kappa = 0, \quad (6.3)$$

at every real Fourier frequency and every positive time.

The narrow correlation width quantifies local persistence around the fixed public-data correlation. It is not a broad calibration-box theorem. The complete closed cover and derivative/direct fallback records are in Appendix B.2; derivative bounds apply to the compactified quantities. The independent reference route of Section 4 does not use this Padé sign proof as its residual certificate.

7. Matched decisions and scientific controls

7.1. Fixed contracts, units and complete ablation

The original six-month contract uses twelve strikes $K_i = 3700 + 100i$, $0 \leq i \leq 11$, forward $F = 4221.86$, discount $D = 1$, and the fixed forward-variance curve in (2.4). Prices and task errors are in index points. The position multiplier is one; no exchange-specific currency notional is inferred. Candidate changes affect α while leaving the curve and other parameters fixed. The three-candidate profile .52, .60, .90 is a finite comparison and excludes several original bid/ask bands; it is not a successful market calibration or a continuous optimum.

The complete $K = 4400$ minus $K = 4500$ task bound is compared below. Each row retains its actual output, signed centre, included finite frequencies, strip, true tail and arithmetic. Displayed upper bounds are rounded upward. The quarter-year study recomputes its maturity-specific reference integral, true-transform envelopes and tails; its upstream full-history certificate is lawfully restricted to the shorter horizon. It does not reuse a six-month exponent or tail value.

Maturity and stage	Complete joint bound, points	Matched signed marginal bound, points	Interpretation
H0: Six months: global state propagation	1.397612096	-	Original failed one-point task
H1: Six months: finite history, same global residual	0.378598956	-	Principal propagation improvement; unchanged output and centre
H2: Six months: local envelope, original used nodes	0.367258782	-	About 3% further bound reduction
H3: Six months: all finite high nodes, global envelope	0.132245064	0.158585551	High-frequency certification and centre change are included
H4: Six months: all finite high nodes, local envelope	0.115215934	0.137896018	Both methods certify 0.25 points
Q0: Three months: through 64, global envelope	1.207557898	1.545191542	Failed 0.25-point task
Q1: Three months: through 128, global envelope	0.351318692	0.408293698	Failed 0.25-point task
Q2: Three months: through 128, local envelope	0.233318843	0.252393939	Only the joint method certifies 0.25 points

The first finite-history reduction is about 72.91%, calculated against the displayed old complete bound. It is a reduction of a guaranteed error bound, not an observed reduction of true pricing error or trading loss. The six-month high-frequency refinement legally changes the centre from approximately -0.166762218 to -0.081698202 points. Its improvement therefore cannot be assigned wholly to common-error geometry. The last row compares equal output, centre, node radii and remainders, and isolates a

decision changed by retaining the shared error variables. Its time-local refinement was fixed after the failed global stage and is an exploratory transfer experiment, rather than a retrospectively preregistered result.

7.2. Twenty-eight portfolio directions

The following matched pass-count table uses H1 at each stated alpha.

Let e_i select strike K_i . We keep the exact holdings below, without rescaling them to make a budget pass. Gross weight means $\sum_i |w_i|$. Every threshold is an absolute error in index points for the specified holding vector.

Family	Exact direction	Count	Gross weight
Adjacent spreads	$e_i - e_{i+1}, 0 \leq i \leq 10$	11	2
Adjacent butterflies	$e_i - 2e_{i+1} + e_{i+2}, 0 \leq i \leq 9$	10	4
Wide spreads	$e_0 - e_3, e_3 - e_6, e_6 - e_9, e_9 - e_{11}$	4	2
Positive baskets	$\frac{1}{12} \sum_{i=0}^{11} e_i, \frac{1}{6} \sum_{i=0}^5 e_i, \frac{1}{6} \sum_{i=6}^{11} e_i$	3	1

All methods in each matched comparison use identical upstream radii. At a 0.5-point budget and $\alpha = .52$, finite-history joint bounds certify 28/28 portfolios against 13/28 signed marginal bounds. At a 0.25-point budget, the corresponding counts are 10/28 against 1/28 for $\alpha = .60$, and 20/28 against 13/28 for $\alpha = .90$. At the looser one-point budget for .52, both new methods certify 28/28; that improvement alone cannot isolate the shared structure. A lawful payoff intersection produces no additional tightening in the tested setting. Failure to certify means that this outer bound is insufficient, not that the true error exceeds the threshold.

7.3. Prespecified nearby candidates under the original quotes

The two objective columns refer to N1 and N2 respectively.

The original twelve half-year quotes, forward $F=4221.86$, discount $D=1$, and all other model parameters are fixed. Before computing new results we froze $\alpha=-0.520, 0.525, 0.530, 0.540, 0.550$, a first layer $N_t=1024$, and an upgrade of **all five** candidates to $N_t=2048$ if any adjacent joint comparison remained unresolved. Every point has a newly generated continuous reference field and a complete closed-time residual bank. Fourier nodes through 64, all omitted finite nodes through 128, true infinite tails, strip remainder, reference arithmetic and original quote-conversion intervals are paid.

The objective is the original normalized midpoint loss $J = \frac{1}{24} \sum_i (c_i - m_i)^2$, using normalized calls. Index-point squared units would require multiplication by F^2 . Here the tiny target interval halfwidth is only outward arithmetic error in converting the fixed bid/ask-price midpoint; it is not the market bid/ask halfwidth. Ranking does not establish a uniform winner for arbitrary quote targets inside those market bands. A Taylor support enclosure preserves common Fourier disks in its linear term and charges the full coordinate-radius quadratic remainder. Its same-radius marginal comparison

uses the identical upstream certificate. Errors across different alpha candidates are not assumed jointly correlated.

alpha	N _t =1024 joint J $\times 10^8$	N _t =2048 joint J $\times 10^8$
0.520	[4.825094, 6.778350]	[5.293449, 6.024095]
0.525	[5.645378, 7.501414]	[6.097920, 6.798485]
0.530	[6.748974, 8.517584]	[7.187004, 7.859529]
0.540	[9.807435, 11.422182]	[10.218707, 10.839486]
0.550	[14.000658, 15.482012]	[14.387007, 14.960680]

Generated layer	Joint strictly separated pairs	Matched marginal pairs
N _t =1024	7/10	3/10
N _t =2048	10/10	7/10

The first layer retained its unresolved close pairs and triggered the frozen all-candidate upgrade. The final finite grid has $\alpha=0.520$ as a strict minimum. This is a five-point comparison, not a continuous calibration optimum. All five candidates remain incompatible with at least one original bid/ask row; the finite-set result therefore does not identify a market-calibrated model.

Both layers fully generate every closed-cell residual entry: 15,754,230 entries, 5,130 reference exponents and 10,250 true-transform envelopes in total. Separate readers check complete coverage and maxima, reconstruct all exponents, coefficients and objectives, and reject deliberate corruptions; they share the stated rigorous primitives rather than independently rederive every residual derivative.

7.4. Why certify a frozen fast output?

The detailed matched output table uses N2L; the preceding failed levels are N1 and N2.

The use case is validation of a stored or externally fixed output: an existing pricing-library audit, a reproducible historical result, or a production output that cannot be replaced in the task being checked. When the reference is already available and output replacement is allowed, directly returning its certified centre is an appropriate control. Centre-correcting the fast output is another control and must include the rounding of both the stored correction and final addition. The reference route does not automatically become an economical online solver merely because its downstream support function is inexpensive.

A separately frozen descriptive control reuses the complete $N_t = 2048, \alpha = .52$ residual bank with 128 history bins. Every intersected closed source cell contributes to each bin maximum, and all weights are outwardly enclosed. All five methods below use the same model, six-month 4400 – 4500 spread, 0.25-point tolerance, reference centre, node radii, strip and true-tail budget. This control does not change the prespecified nearby-grid experiment.

Output	Complete joint, points	Matched marginal, points	Joint 0.25-point decision
Frozen Pad�	0.394999331	0.514096070	UNRESOLVED
Direct reference (binary64)	0.228237113	0.347333852	PASS
Fast + stored correction	0.228237113	0.347333852	PASS
BL core, 512 steps	0.229082759	0.348179497	PASS
BL core, 1024 steps	0.228534686	0.347631424	PASS

All five matched marginal bounds remain above 0.25 points. Direct reference and corrected fast values happen to have the same complete bound in this instance; their actual stored values are checked separately. Exact dyadic arithmetic charges the reference return, stored correction and final addition. For these freely replaceable outputs, the reference is simpler than delivering the unchanged fast result. The joint structure still changes the decision for the reference and BL outputs. The BL 512-to-1024 difference is a diagnostic, not its certified error.

The certificate is not free. Its generation encloses 2,100,735 continuous frequency-by-cell residual entries, 513 reference exponents and 1,025 true-transform node bounds, plus the infinite tail and 12,300 price coefficients. Full reading checks all these entries and reconstructs the exponents, coefficients and decisions. Each additional portfolio evaluates 1,025 shared-disk support terms and the complete remainder; 28 portfolios reuse the same bank without further residual generation. This is reuse across directions at one parameter and maturity, not reuse across unverified model parameters.

Additional output construction	Deterministic mathematical work
Original Pad� output	352 Fourier values; 256 Jacobi samples per frequency
Direct returned reference	Return the already generated, fully charged centre with binary64 rounding
Corrected fast output	Store one correction and perform one binary64 addition per price
BL core, 512 steps	67,371,264 history scalar-vector products; 2,626,560 transformed Riccati evaluations
BL core, 1024 steps	269,222,400 history scalar-vector products; 5,253,120 transformed Riccati evaluations

These are typed work counts, not interchangeable floating-point operations or a total-cost ratio. The field generation, outward transcendental evaluations and verification remain distinct from nominal output construction. No speed or machine-performance advantage is inferred. Complete ledgers record the additional exponent-quadrature work and both coarse failed control stages.

7.5. What the experiments do and do not establish

The modern-method comparison reimplements the modified-Adams Riccati core described by Boyarchenko et al.; it does not reproduce their full SINH deformation or Conformal Bootstrap. Agreement between two discretizations is a numerical diagnostic, not an interval certificate. Any comparison to a complete certified output must pay the independent reference, tails, arithmetic and verification costs required by that guarantee.

The fixed refinement menu certifies reliable termination when the complete task radius meets the budget. Its minimum number of actions assumes that every alternative radius is already certified and every action has unit cost. We retain those assumptions and do not infer minimum work for an unknown menu. Work spent constructing all alternatives remains part of certificate generation. Model fit, market uncertainty, transaction costs and global continuous calibration remain outside the output-error guarantee.

7.6. Matched dissipative-kernel evaluation

We froze the six-month configuration at $\alpha = 13/25$, $\nu = 2897/10000$, $\rho = -1489/2000$, and $\sigma = 1489/4000$ before computing the new resolvent weights. The physical damping is therefore $\lambda = \nu\sigma = 4313633/40000000$. All four comparisons reuse the same N2L (N_t=2048) closed-cell residual bank, the same stored binary64 fast prices, the same reference exponents and signed centres, and the same finite omission, analytic strip, true infinite tail and reference arithmetic terms. The financial direction is the unit 4400-4500 call spread; the full finite grid contains 1025 frequencies, including 513 used reference frequencies and 512 retained finite omitted frequencies. The complete twelve-price certificate is reconstructed for every method.

The table reports upper endpoints in index points, rounded upward to nine decimals. "Global state" first bounds the state by its global residual and then applies the exponent functional; "global curve" propagates that same residual maximum directly through the curve. The final two rows use the identical 128 binwise envelopes and differ only in the pricing kernel.

Propagation	Frozen fast marginal	Frozen fast joint	Stored reference joint	Used-node joint radius
Global state	3.690877564	1.829428107	1.662665889	1.476801373
Global curve	0.589267065	0.428570987	0.261808769	0.075944253
Local curve	0.514096070	0.394999331	0.228237113	0.042372598
Local dissipative resolvent	0.509733613	0.393082828	0.226320610	0.040456094

The actual binary64 corrected-fast output has the same certified bounds as the stored reference in this configuration; its conversion and rounding are included in the read-back. The strict rational reduction from local curve to local resolvent is positive and approximately 0.001916504 points: about 0.485% of the complete frozen-fast bound and 4.523% of its used-node joint radius. These percentages describe changes in guaranteed upper bounds, rather than observed pricing error or transaction loss. The

one-point and half-point frozen-output tasks remain certified; the quarter-point frozen-output task remains unresolved. Both local methods already certify the stored reference and corrected output at a quarter point. Thus the additional dissipative kernel does not change that budget decision in this matched experiment.

The modest gain is consistent with the fixed terms in the certificate. Even if the contribution of all 513 used reference nodes vanished, the same symmetric frozen-output budget formula would retain the absolute signed centre, finite omitted-node radius and other remainder terms. Their exact rational sum is strictly greater than the downward decimal endpoint 0.352626733 points, and is therefore greater than a quarter point. This is a floor for the declared fixed budget formula, not a lower bound on the true error, and not a floor for other certification strategies. Keeping those terms frozen prevents a quarter-point certificate through refinement of the used nodes alone. The omitted finite frequencies and the output-centre conversion therefore remain the consequential next research targets.

The reconstructed global-curve and local-curve endpoints agree exactly with the saved baseline rational endpoints, including all twelve individual radii and frozen-output bounds. For example, the saved global-curve endpoint is approximately 0.4285709861017229; its strict nine-decimal upward display is 0.428570987. A previously rounded display of 0.428570986 is not a change in residuals, centres, coefficients or remainders.

Verification scope and mathematical work

The separate reader actually reads all 2,100,735 entries of the complete residual bank, checks closed-cell coverage from zero, and recomputes the maxima for all 128 propagation bins. It independently expands the cumulative resolvent by the direct double Gamma series with fourteen outer terms, while the producer uses twelve outer terms and a strict power-field moment. Both enclose all inner tails with sixty-four retained terms. The reader checks all 513 transform-radius minima in all four comparisons and rebuilds all twelve prices with all 1025 finite frequencies. At $\sigma = 0$, it replays all 129 cumulative endpoints and all 128 cell weights against the curve-kernel conclusion. Eight actual negative controls are rejected: negative damping, an incorrect physical ν scale, a missing initial closed cell, a removed finite omitted node, an erased true infinite tail, an altered signed centre, an understated resolvent exponent enclosure, and a zero-damping weight that does not reproduce the curve kernel.

This is an independent reconstruction of the new cumulative-weight algebra and certificate assembly. It shares the identified dyadic arithmetic, Gamma, logarithm and exponential primitives, and inherits the previously validated residual derivative proof and reference exponents. It is not an independent rederivation of those upstream proofs. The execution receipts distinguish that sharing boundary explicitly.

The new propagation reuses the existing bank and generates zero new residual entries. Its declared weighted residual sum has $513 \times 128 = 65,664$ terms, with 129 cumulative endpoint enclosures, twelve outer series terms per producer endpoint, and sixty-four inner curve terms. These are typed mathematical work counts. They do not establish a total computational cost, a speedup, or comparability with workloads that regenerate

residuals or reference prices. The frozen contract, strict outputs and separate reader are supplied in the kernel evidence component.

7.7. Matched ledgers and certificate resolution

The quarter-maturity decision is attributable to shared aggregation. Q2 uses identical actual output, reference centre, 1025 node radii and all remainders in both columns. Finite omitted-node support is zero because every finite node has a nonzero reference. Values below are in index points; exact fractions are retained in quarter-exact-ledger.json.

Component	Joint	Signed marginal
Signed centre	-0.149775146827	-0.149775146827
Absolute centre charge	0.149775146827	0.149775146827
Used-node support	0.045461793432	0.064536889507
Finite omitted-node support	0.000000000000	0.000000000000
Strip remainder	0.000026654230	0.000026654230
True infinite-tail remainder	0.038055248032	0.038055248032
Reference arithmetic	4.148871871727e-16	4.148871871727e-16
Complete upper bound	0.233318843	0.252393939

The joint complete bound is 0.233318843 points versus 0.252393939 points for signed marginal aggregation; only the joint certificate passes the 0.25-point budget. The difference 0.019075095 points is entirely the node-support aggregation difference. Complete-bound displays are rounded upward to nine decimal places; component/centre and gap displays are approximate. Every decision and ledger identity uses exact fractional endpoints. Tiny reference arithmetic is displayed separately, and the signed centre is a locator, not an extra additive charge on top of its absolute value.

Every portfolio curve uses the 28 originally fixed and fully specified directions. Exact endpoints, all threshold breakpoints and pass counts are supplied in portfolio-exact-thresholds.csv and portfolio-budget-counts.json. Certification uses the complete rational endpoint condition $B \leq \tau$, including equality. Between-bank plots show descriptive changes; joint versus signed marginal within one bank uses matched radii. No 28-direction quarterly dataset is inferred from the single quarter spread.

Bank	alpha	Budget points	Marginal	Joint
H0	13/25	1/4	0/28	0/28
H0	13/25	1/2	0/28	0/28
H0	13/25	1	0/28	3/28
H1	13/25	1/4	0/28	2/28
H1	13/25	1/2	13/28	28/28

Bank	alpha	Budget points	Marginal	Joint
H1	13/25	1	28/28	28/28
H1	3/5	1/4	1/28	10/28
H1	3/5	1/2	22/28	28/28
H1	3/5	1	28/28	28/28
H1	9/10	1/4	13/28	20/28
H1	9/10	1/2	25/28	25/28
H1	9/10	1	27/28	28/28
N1	13/25	1/4	0/28	0/28
N1	13/25	1/2	0/28	4/28
N1	13/25	1	15/28	28/28
N2	13/25	1/4	0/28	0/28
N2	13/25	1/2	6/28	23/28
N2	13/25	1	28/28	28/28
N2L	13/25	1/4	0/28	1/28
N2L	13/25	1/2	11/28	26/28
N2L	13/25	1	28/28	28/28

The two nearby levels retain all ten unordered pairs, including unresolved ones:

N't	Method	Separated pairs	Complete unresolved list
1024	joint	7/10	(0.520, 0.525), (0.520, 0.530), (0.525, 0.530)
1024	marginal	3/10	(0.520, 0.525), (0.520, 0.530), (0.520, 0.540), (0.525, 0.530), (0.525, 0.540), (0.530, 0.540), (0.540, 0.550)
2048	joint	10/10	none
2048	marginal	7/10	(0.520, 0.525), (0.520, 0.530), (0.525, 0.530)

The closest N2 joint separation, alpha=.520 versus .525, is a positive gap of approximately $7.382643478687e-10$ in normalized squared-midpoint loss. Its endpoint identity is $J_{0,B} - J_{0,A} - H_B - H_A - Q_A$ plus any box-intersection endpoint adjustment. The gap decomposition, including used nodes, finite zero nodes, reference strip/tail/arithmetic and midpoint-conversion arithmetic, is recorded exactly in nearby-pair-resolution.json. This is a fixed original midpoint-loss ranking. The midpoint-conversion arithmetic interval does not represent the market bid/ask width; all five candidates remain separately bid/ask-incompatible.

The unchanged actual half-year 4400-4500 output admits the following complete centre accounting. Signed centres are approximate displays; radius and complete-bound columns are decimal upper endpoints, so independently rounded columns need not add exactly:

Bank	Signed centre points	Complete radius points	Complete bound points
H2	-0.166762218	0.200496564	0.367258782
H3	-0.081698202	0.050546863	0.132245064
H4	-0.081698202	0.033517732	0.115215934
N2	-0.166762218	0.261808769	0.428570987
N2L	-0.166762218	0.228237113	0.394999331

Expanding H2 to H3 changes the absolute centre charge as well as the paid reference radius; H3 to H4 retains that new centre. N2 to N2L retains its own reference centre and fullremainders. H4 to N2L is a descriptive cross-bank identity, not an ablation. These transitions are exact in frozen-output-centre-account.json, which also reports the common strict-reference certificate floor as a fraction of each BL-core bound. A dominant floor limits what this comparison can establish about intrinsic solver accuracy. When replacement is permitted and the strict reference is already available, returning that reference directly is the simpler workload; frozen-output audit and free replacement answer different tasks.

8. Applicability and a fixed reproducibility entry

The guarantee is conditional on the stated model, affine transform, regular reference and outward enclosures. The analytical condition on the curve is not itself a stochastic admissibility theorem. The narrow structural domain excludes general mean reversion and broad correlation calibration. Unresolved intervals, failed budgets and quote incompatibilities are valid outcomes.

The fixed release `v3.0.0-research-20261007` links this paper, editable sources, the scientific evidence and commands for the same revision. The named evidence archive contains the numerical banks; GitHub’s automatically generated source archive is not a substitute. File checksums prove identity. Scientific readers establish only the mathematical obligations that their code actually recomputes.

The verification matrix in Appendix E assigns each obligation to a generation command, reading command, recomputation scope and shared dependency. In particular, `--full` is not an assertion that every continuous derivative enclosure has been independently regenerated. It adds the specified structural coverage and reconstruction checks. Residual regeneration has separate explicit commands. The readers reconstruct history maxima, strict exponents, transform and coefficient enclosures, complete prices, objectives and decisions where listed; saved derivative bounds and common strict primitives remain part of the trusted base. An author-side fresh execution is not an external referee reproduction.

Software versions and mathematical workload counts are sufficient to identify the numerical procedure, without personal host metadata or elapsed measurements. Counts of closed cells, interval terms, history products and direction supports have different units. They establish the amount of work performed and what is reused, rather than a speed comparison. Evidence volume is disclosed separately. A new direction at fixed

parameter and maturity reuses the same certified bank; the generation cost cannot be omitted from a claimed full task.

Appendices A-E complete the main proof and experiments. Appendices F-G are independent classical Heston extensions with original-chain probability objects and a four-part signed identity for inexact trial functions. They do not assert a completed annual monetary pricing certificate. The earlier thirteen-dimensional receipt lacks the full coefficient bank required for that claim.

The finite-history theory and matched task experiments establish a bounded, reusable route from continuous residuals to actual-output certificates. Direct certified reference output should be used when replacement is permitted and available; fixed-output audit and shared-bank reuse are distinct workloads. No point estimate is treated as exact truth, and no certificate-to-true-error ratio is reported when a rigorous error interval crosses zero.

Appendix A. Verification of the Probability-Model and Strip-Transform Assumptions

The primary source is Abi Jaber-El Euch, arXiv:1803.00477v1, first posted on 2018-03-01, with a PDF title-page date of March 2, 2018. The journal article appeared in 2019 in Statistics & Probability Letters 149, 63-72. Locators refer to the internal pagination of the 13-page preprint: Theorem 2.1 and Example 2.2 on p.4, Theorem 2.3 on p.5, and H2 and Table 1 on pp.9-10. These are preprint, not journal, page numbers.

For $K_\alpha = t^{\alpha-1}/\Gamma(\alpha)$,

$$\int_0^h K_\alpha^2 dt = \frac{h^{2\alpha-1}}{(2\alpha-1)\Gamma(\alpha)^2}, \quad \int_0^T (K_\alpha(t+h) - K_\alpha(t))^2 dt \leq \frac{h^{2\alpha-1}}{\Gamma(\alpha)^2} \int_0^\infty [(r+1)^{\alpha-1} - r^{\alpha-1}]^2 dr. \quad (\text{A.1})$$

The exponent in the latter integral is $2\alpha-2 > -1$ at zero and $2\alpha-4 < -1$ at infinity, so the integral is finite. The resolvent of the first kind,

$$\mathcal{L}_\alpha(dt) = t^{-\alpha} dt / \Gamma(1-\alpha), \quad K_\alpha * \mathcal{L}_\alpha = 1 \quad (\text{A.2})$$

is verified by the Beta identity and is nonnegative and nonincreasing. The completely monotone spectral measure

$$\mu_\alpha(dx) = x^{-\alpha} dx / [\Gamma(\alpha)\Gamma(1-\alpha)] \quad (\text{A.3})$$

gives $K_\alpha(t) = \int e^{-xt} \mu_\alpha(dx)$. Under $r = xh$, the two spectral integrals in H2 become constants times $h^{\alpha-1}$ and $h^{\alpha-1/2}$, respectively. The remaining constants $\int (1 \wedge r^{-1/2}) r^{-\alpha} dr$ and $\int r^{-\alpha-1/2} (1 \wedge r) dr$ are finite for $.5 < \alpha < 1$. Table 1 of the source therefore covers the required shifted-kernel conditions.

Curve (2.4) is nondecreasing, has a nonnegative initial value, and is locally Hölder with exponent $\alpha_0 = .5286$. On $\alpha \in [.52, .9]$, the kernel has $\gamma/2 = \alpha - .5 \leq .4 < \alpha_0$. Hence the curve-class assumptions in Example 2.2(i) hold, and Theorem 2.1 gives a nonnegative continuous weak variance solution and moment bounds. The equivalent kernel input is

$$b_\alpha(t) = D_C^\alpha \xi_*(t) = I^{1-\alpha} \xi'_*(t) \geq 0, \quad b_\alpha(t) = \frac{(\theta - V_0)\lambda_\xi}{\Gamma(1 + \alpha_0 - \alpha)} t^{\alpha_0 - \alpha} + O(t^{2\alpha_0 - \alpha}) \quad (\text{A.4})$$

Its worst exponent is $-0.3714 > -0.5$, so it belongs to L_{loc}^2 ; also $\xi_* = V_0 + K_\alpha * b_\alpha$. The nonnegative input measure meets Example 2.2(ii). This verifies the model specification with a varying kernel and fixed forward variance curve.

Theorem 2.3 of the source requires $\text{Re } \psi_1 \in [0, 1]$, a second-state initial transform with nonpositive real part, and a variance forcing term with nonpositive real part. Taking $\psi_1 = ia = 1/2 + iu$ and zero for the other initial and forcing terms gives (2.2), (2.6), and uniqueness in weak law. With $\psi_1 = 1$, the Riccati solution is zero and $\mathbb{E}S_T = S_0$ follows. A positive local martingale with constant expectation is a true martingale. Thus the probability strip in (4.2) follows from the continuous-time model. This argument applies established existence and affine-transform theory.

Appendix B. Exact construction and structural certificates

B.1. Matching algebra and full-frequency proof

Set $S = s_0 - i\rho u = -d$, take the principal square root in $A = \sqrt{S^2 + 2b}$, and let $R = A - S$. The three short-time and long-time coefficients are respectively

$$b_1 = -\frac{b}{\Gamma(1 + \alpha)}, \quad b_2 = \frac{Sb}{\Gamma(1 + 2\alpha)}, \quad b_3 = \frac{\Gamma(1 + 2\alpha)}{\Gamma(1 + 3\alpha)}(db_2 + b_1^2/2), \quad (\text{B.1})$$

$$g_0 = -R, \quad g_1 = \frac{R}{A\Gamma(1 - \alpha)}, \quad g_2 = -\frac{R}{A^2\Gamma(1 - 2\alpha)} + \frac{R^2}{2A^3\Gamma(1 - \alpha)^2}. \quad (\text{B.2})$$

For $1/2 < \alpha < 1$, $\Gamma(1 - 2\alpha)$ is finite and negative. The fixed two-endpoint construction of Gatheral and Radoičić [GR2019] requires

$$\begin{aligned} \hat{H}(y) &= P(y)/Q(y), \quad Q = 1 + q_1y + q_2y^2 + q_3y^3, \\ \hat{H} &= b_1y + b_2y^2 + b_3y^3 + O(y^4), \quad \hat{H} = g_0 + g_1y^{-1} + g_2y^{-2} + O(y^{-3}). \end{aligned} \quad (\text{B.3})$$

These six conditions give the linear system

$$\begin{pmatrix} g_0 & g_1 & g_2 \\ b_1 & -g_0 & -g_1 \\ b_2 & b_1 & -g_0 \end{pmatrix} \begin{pmatrix} q_1 \\ q_2 \\ q_3 \end{pmatrix} = \begin{pmatrix} b_1 \\ -b_2 \\ -b_3 \end{pmatrix}, \quad p_1 = b_1, \quad p_2 = b_2 + b_1q_1, \quad p_3 = g_0q_3. \quad (\text{B.4})$$

Equations (B.1)-(B.4) are the established construction. We next prove invertibility, a nonvanishing denominator, the trajectory half-plane property, and exact-solution error bounds for an independent reference trajectory.

To avoid division by the matching determinant before proving its nonvanishing, define

$$f = \Gamma(1 + \alpha)^{-1}, \quad p = \frac{\sin \pi\alpha}{\pi\alpha}, \quad v = -\cos \pi\alpha, \quad m_\alpha = \frac{\Gamma(1 + 2\alpha)}{\Gamma(1 + \alpha)^2}, \quad \zeta = \frac{\Gamma(1 + 2\alpha)\Gamma(1 + \alpha)}{\Gamma(1 + 3\alpha)},$$

$$M_2 = Sb/m_\alpha, \quad c = \zeta(b^2/2 - S^2b/m_\alpha), \quad V = pR/A, \quad W = m_\alpha p v R/A^2 + p^2 R^2/(2A^3). \quad (B.5)$$

The reflection formula and recurrence relations give $b_1 = -fb, b_2 = f^2 M_2, b_3 = f^3 c$ and $g_1 = V/f, g_2 = W/f^2$. In the latter two expressions, the factors f, f^2 occur in the denominators. Direct expansion of the determinant and Cramer numerators of (B.4) gives

$$\begin{aligned} \Delta &= b^2 W + 2bRV - M_2 RW - M_2 V^2 - R^3, \\ F_1 &= b^2 V + bM_2 W - bR^2 + M_2 RV + cRW + cV^2, \\ F_2 &= -b^2 R + bM_2 V + bcW + M_2^2 W + M_2 R^2 + cRV, \\ F_3 &= -b^3 + 2bM_2 R - bcV - M_2^2 V + cR^2. \end{aligned} \quad (B.6)$$

Here Δ is exactly the determinant of the original system, and the numerators are $N_j = f^j F_j$. After establishing $\Delta \neq 0$, we may define $q_j = N_j/\Delta$, in which case

$$\Delta P = -fb\Delta y + f^2(M_2\Delta - bF_1)y^2 - f^3 R F_3 y^3, \quad \Delta Q = \Delta + fF_1 y + f^2 F_2 y^2 + f^3 F_3 y^3. \quad (B.7)$$

In physical time, the denominator coefficients are $\nu^j q_j$; the normalised coefficients q_j retain their original definition.

Proof. First take $u \geq 0$. Set

$$\begin{aligned} \omega &= \sqrt{u^2 + 1/4}, \quad \eta = u/(1+u), \quad j(\eta) = \sqrt{\eta^2 + (1-\eta)^2/4}, \quad s = -\rho/2, \\ \bar{S} &= s(1-\eta+2i\eta)/j(\eta), \quad \bar{A} = \sqrt{1+\bar{S}^2}, \quad \bar{R} = (\bar{A} + \bar{S})^{-1}. \end{aligned} \quad (B.8)$$

Since $j^2 \geq 1/5$, these functions are defined on the closed interval $\eta \in [0, 1]$. We have $|\bar{S}| = |\rho|$ and

$$\text{Re } \bar{A}^2 = 1 - \rho^2 + 2(\text{Re } \bar{S})^2 \geq 1 - \rho^2 > 0.$$

The principal square root is continuous and lies in the first quadrant; $\text{Re } \bar{A}, |\bar{A}| > 3/5$. The real part of the Hermitian product of two first-quadrant numbers is nonnegative. Hence

$$|\bar{A} + \bar{S}|^2 \geq |\bar{A}|^2 + |\bar{S}|^2 \geq |\bar{A}^2 - \bar{S}^2| = 1, \quad |\bar{R}| \leq 1, \quad \text{Re } \bar{R} > 0.$$

Equation (B.8) expresses $\bar{A} - \bar{S}$ in reciprocal form, with a denominator determined directly by the current parameters.

Substitute $\bar{b} = 1/2$ and (B.8) into (B.5)-(B.6) to obtain the barred original quantities. Homogeneity gives

$$\Delta = \omega^3 \bar{\Delta}, \quad F_j = \omega^{3+j} \bar{F}_j. \quad (B.9)$$

Define three real functions

$$\bar{B}_j = \text{Re}(\bar{F}_j \bar{\Delta}). \quad (B.10)$$

The rigorous rational covering in Appendix B proves that, throughout the closed rectangle $[13/25, 3/5] \times [0, 1]$, $\bar{B}_j > 0$ and

$$|\bar{\Delta}|^2 \geq \frac{88110801209184778874628745}{1267650600228229401496703205376} > \frac{1}{14400}. \quad (\text{B.11})$$

This first establishes $\Delta \neq 0$ and then yields $\text{Re } q_j = (f\omega)^j \bar{B}_j / |\bar{\Delta}|^2 > 0$, proving (6.2).

For the trajectory half-plane, the finite convolution in (B.7) gives

$$|\Delta|^2 \text{Re}(P\bar{Q}) = \sum_{n=1}^6 f^n D_n y^n, \quad (\text{B.12})$$

$$\begin{aligned} D_1 &= -b|\Delta|^2, \\ D_2 &= \text{Re}(\mathbf{M}_2)|\Delta|^2 - 2b \text{Re}(F_1 \bar{\Delta}), \\ D_3 &= \text{Re}\{-RF_3 \bar{\Delta} + (\mathbf{M}_2 \Delta - bF_1) \bar{F}_1 - b\Delta \bar{F}_2\}, \\ D_4 &= \text{Re}\{-RF_3 \bar{F}_1 + (\mathbf{M}_2 \Delta - bF_1) \bar{F}_2 - b\Delta \bar{F}_3\}, \\ D_5 &= \text{Re}\{-RF_3 \bar{F}_2 + (\mathbf{M}_2 \Delta - bF_1) \bar{F}_3\}, \\ D_6 &= -\text{Re } R |F_3|^2. \end{aligned} \quad (\text{B.13})$$

The same rational covering certifies the six compactified quantities $\bar{D}_n < 0$, with scaling $D_n = \omega^{7+n} \bar{D}_n$. Set $z = f\omega y$. The right-hand side of (B.12) is $\omega^7 \sum_{n=1}^6 \bar{D}_n z^n$ and is therefore strictly negative for every $y > 0$. Having proved that the denominator is nonzero, division by $|Q|^2 |\Delta|^2$ gives $\text{Re } \hat{H} < 0$. Negative frequencies follow by conjugate symmetry of the principal root and matching coefficients. The closed endpoint $\eta = 1$ represents the infinite-frequency limit, so the proof on the closed compactified domain covers every finite frequency. ■

Positive real parts of the coefficients provide a sufficient condition for a nonvanishing denominator. Appendix B specifies the algebraic identities, elementary-function remainder bounds, and continuous interval covering used in the computer-assisted proof.

A continuous, quantitatively bounded correlation extension

The original domain was chosen to contain the public-data example's correlation and roughness candidates while keeping the six-condition construction fixed. The sign cover established a continuous roughness interval and all frequencies; it was not intended as a theorem over typical calibration boxes. Strict signs imply persistence in correlation. The following result makes that implication quantitative without replacing a continuous cover by a sampled grid.

The certificate domain is $\mathcal{B} = [13/25, 3/5] \times [0, 1]$. Every interval $I = [\ell/2^{100}, r/2^{100}]$ uses integer outward rounding. Rational inputs are rounded down and up; products take extrema over the four endpoint corners and are quantised. Reciprocals first exclude zero. Square roots use integer isqrt and an increment for the upper endpoint. Complex quantities use rectangular real-imaginary interval arithmetic. The analytic lower bounds in (B.8) fix the principal-root and reciprocal branches; no floating-point sign tolerance is used.

The four basic α -dependent functions p, v, m_α, ζ satisfy

$$p' < 0, \quad v' > 0, \quad (\log m_\alpha)' = 2[\psi(1 + 2\alpha) - \psi(1 + \alpha)] > 0,$$

$$(\log \zeta)' = 2[\psi(1 + 2\alpha) - \psi(1 + 3\alpha)] + [\psi(1 + \alpha) - \psi(1 + 3\alpha)] < 0. \quad (\text{B.14})$$

We have $\psi'(x) = \sum_{n \geq 0} (x + n)^{-2} > 0$, and elementary trigonometric identities give the derivatives of p, v . Strict endpoint values therefore enclose each continuous α interval, without sampling α at grid points.

The constant π is certified by Machin's identity and one hundred terms of each alternating arctangent series. The logarithm is reduced to $[1, 2]$ and evaluated with one hundred positive atanh-series terms and a geometric tail. The exponential is reduced to $[0, 1]$, evaluated with one hundred Taylor terms and a geometric tail, and restored by repeated rigorous squaring. For positive real Gamma arguments, recurrence first shifts to $z \geq 20$. Retaining $B_2, \dots, B_{20}$ in log Gamma gives the positive-real remainder bound

$$0 < R_{10}(z) < B_{22}/(22 \cdot 21z^{21}). \quad (\text{B.15})$$

In Binet's positive integral, the ten-term geometric remainder of arctangent is positive and bounded by the first omitted power. Integration gives (B.15), consistent with the positive-real Stirling remainder in NIST2010. After exponentiation, divide successively by the recurrence factors. Sine and cosine retain thirty-two terms, with absolute Lagrange tails $M^{65}/65!$, $M^{64}/64!$. Bernoulli numbers, factorials, and interval endpoints are exact rationals.

For each closed rectangle, substitute these enclosures successively into (B.8), (B.5), (B.6), (B.10), and (B.13). Accept only when the lower endpoints of all three $\bar{B}_j$ are strictly positive and the upper endpoints of all six $\bar{D}_n$ are strictly negative. Otherwise bisect one coordinate at its exact rational midpoint. The two closed children have the parent rectangle as their union, and strict enclosures also cover the shared boundary.

The finite covering contains 211241 leaf rectangles of total rational area exactly $2/25$. An independent check reconstructs 422481 binary-tree nodes from the root, with maximum depth 19 and each recorded leaf occurring exactly once. In addition to the area check, this excludes missing subtrees, interior overlap, and endpoint gaps. The theorem uses the order interval covered by this complete closed tree.

The stronger rational margins in the complete certificate imply the following bounds. Each simplification has been independently checked using Fraction.

Quantity on the full compactified domain	Strict lower bound
$\bar{B}_1, \bar{B}_2, \bar{B}_3$	$1/6000, 1/8000, 1/20000000000$
$-\bar{D}_1, -\bar{D}_2, -\bar{D}_3$	$1/30000, 1/20000000000, 1/6000$
$-\bar{D}_4, -\bar{D}_5, -\bar{D}_6$	$1/8000, 1/15000000, 1/100000$
$ \bar{\Delta} ^2$	$1/14400$

The proof consists of rigorous interval inclusion and a complete finite covering. The supplementary material provides the interval implementation, all leaf rectangles, and the tree-structure check for independent verification.

B.2. Continuous correlation persistence

Proof. Retain the frequency compactification $\eta = u/(1+u)$, $0 \leq \eta \leq 1$, and the original unnormalized polynomials Δ, F_j, B_j, D_j . Write $S_b = -\rho[(1-\eta) + 2i\eta]/(2h)$, $h^2 = \eta^2 + (1-\eta)^2/4$. Then $|S_b| = |\rho|$, $\Re S_b \geq 0$ for negative ρ , $A^2 = 1 + S_b^2$, and $R = (A + S_b)^{-1} = A - S_b$. Throughout $|\rho| \leq 3/4$, the square-root branch remains fixed because $\Re A^2 \geq 1 - \rho^2 > 0$. Consequently,

$$|A|^{-1} \leq 8/5, \quad |A'| \leq 6/5, \quad |(A^{-1})'| \leq 384/125, \quad |R| \leq 1, \quad |R'| \leq 11/5, \quad (\text{B.16})$$

where primes denote real ρ derivatives. The identity $|R| \leq 1$ follows from $|A + S_b|^2 \geq 1$: the terms $\Re A \Re S_b$ and $\Im A \Im S_b$ are nonnegative, while $|A|^2 = |1 + S_b^2| \geq 1 - |S_b|^2$. Also $\Re R > 0$, since $(\Re A)^2 - (\Re S_b)^2 = (|1 + S_b^2| + 1 - |S_b|^2)/2 > 0$.

For the α -dependent scalars of Theorem 6.1, exact outward endpoint enclosures give

$$0 < p \leq 2/3, \quad 0 \leq v \leq 1/3, \quad 1 \leq m \leq 3/2, \quad 0 < \zeta \leq 2/3. \quad (\text{B.17})$$

Here p decreases and v, m increase on the stated interval, while ζ decreases. The gamma-ratio assertions follow from the increasing digamma function; $m \geq 1$ also follows from log convexity of Γ . None depends on ρ .

Apply product-rule modulus bounds to the original polynomial formulas. For every (α, η) , the resulting exact rational constants $L_{B,j}, L_{D,j}$ satisfy

$$|\partial_\rho B_j| \leq L_{B,j}, \quad |\partial_\rho D_j| \leq L_{D,j}. \quad (\text{B.18})$$

They are assembled using pairs (M, N) , meaning $|f| \leq M$, $|f'| \leq N$, with addition $(M_1 + M_2, N_1 + N_2)$ and multiplication $(M_1 M_2, N_1 M_2 + M_1 N_2)$. This yields a finite, directly checkable rational derivation rather than a numerical derivative estimate.

Let $m_{B,j}(C), m_{D,j}(C)$ be the exact original sign lower bounds on a continuous (α, η) cell C at $\rho_0 = -1489/2000$. A cell is certified throughout the target correlation interval whenever all bounds

$$m_{B,j}(C) - 10^{-6} L_{B,j} > 0, \quad m_{D,j}(C) - 10^{-6} L_{D,j} > 0 \quad (\text{B.19})$$

hold. The mean-value theorem proves this sufficient test; it uses a derivative enclosure, not numerical differentiation. It certifies 182002 original leaves and 79758 finer leaves. For the remaining cells, directly evaluate the original complex polynomial formulas with ρ , α and η all interval valued. Exact outward dyadic arithmetic certifies strict $B_j > 0$ and $D_j < 0$ on 19808 additional closed cells.

The original complete midpoint tree and the exact local splice trees verify that these 281568 cells cover the entire (α, η) rectangle without gaps or interior overlap. Their exact area is $2/25$; each cell carries the entire correlation interval, giving exact three-dimensional volume $1/6250000$. Hence the raw signs hold for every point of the stated domain. In particular $\Delta \neq 0$, the normalized denominator coefficients have positive real part, and all real numerator coefficients of the half-plane test are negative. The original algebraic implication gives (6.3), including $\eta = 1$, the infinite-frequency limit. Conjugacy supplies negative frequencies. ■

This is a narrow local robustness result: the total correlation width is 2×10^{-6} . It does not establish a typical broad calibration box or a new price experiment at an altered correlation. The direct interval supplement is explicitly exploratory: its protocol was frozen after the derivative-only attempt left positive unresolved area. The evidence retains that partial result and the separately valid conservative fallback.

The independent reader checks every saved original sign, the original 422481-node partition tree, the local splice geometry, the independently assembled derivative majorants and all 19808 direct cells through separate Cramer matrices and numerator-polynomial convolution. The dyadic primitives and saved original sign generation remain shared dependencies; every old interval-polynomial evaluation is not regenerated. Exact large fractions stay in the machine ledger. Neither a sampled correlation grid nor an assertion of continuity replaces any cell in this proof.

Appendix C. Complete propagation proofs and analytic variants

C.1. Regularity, convexity and dissipative state comparison

If $H = I^\alpha F(H)$ is a bounded local continuous solution, then $H, F(H)$ are α -Hölder continuous. For $\alpha > 1/2$, write $q = F(H)$. Cancellation gives the derivative formula

$$H'(x) = \frac{q(x)x^{\alpha-1}}{\Gamma(\alpha)} + \frac{\alpha-1}{\Gamma(\alpha)} \int_0^x (x-t)^{\alpha-2} [q(t) - q(x)] dt. \quad (C.1)$$

The bound

$$|H'(x)| \leq \frac{\|q\|_\infty x^{\alpha-1}}{\Gamma(\alpha)} + \frac{(1-\alpha)[q]_{C^\alpha} x^{2\alpha-1}}{\Gamma(\alpha)(2\alpha-1)}. \quad (C.2)$$

is integrable at zero. Positive-time continuity and the limit from truncated intervals establish absolute continuity on the closed interval. The near-endpoint exponent satisfies $2\alpha-2 > -1$; the other endpoint and the first term are integrable. Thus $H \in AC$ and it is locally C^1 at positive times. Local existence follows from Volterra contraction on a bounded ball. For $v \in AC$ that is locally Lipschitz at positive times, integration by parts gives

$$D_C^\alpha v(x) = \frac{1}{\Gamma(1-\alpha)} \left\{ \frac{v(x) - v(0)}{x^\alpha} + \alpha \int_0^x \frac{v(x) - v(t)}{(x-t)^{1+\alpha}} dt \right\}. \quad (C.3)$$

At a positive maximum over the full history, with zero initial value, this derivative is strictly positive. Consequently, $D_C^\alpha v + sv \leq D_C^\alpha w + sw$, $v(0) = w(0)$, and $s \geq 0$ imply $v \leq w$. For a convex C^1 function Φ on the real plane, apply the supporting-hyperplane inequality to both terms in (C.3) to obtain

$$D_C^\alpha \Phi(v) \leq \nabla \Phi(v) \cdot D_C^\alpha v. \quad (C.4)$$

This is the established Caputo history-convexity inequality, rather than an ordinary chain rule. We work within the regularity assumptions of Proposition 3.11 of Li and Liu [LiLiu2018] and directly prove the version needed here.

Lemma C.1. Suppose $\alpha \in (1/2, 1)$, $|\rho| \leq 1$, $s_0 \geq 0$. The normalised equation has a unique global solution, and $\operatorname{Re} H(x) < 0$ for $x > 0$. If $s_0 > 0$, then

$$|H(x)| \leq \frac{b}{s_0} [1 - E_\alpha(-s_0 x^\alpha)] \leq \min\{b/s_0, bx^\alpha/\Gamma(1+\alpha)\}. \quad (\text{C.5})$$

When $s_0 = 0$, the latter time-dependent bound remains valid.

Proof. Write $H = X + iY$. Completing the square gives

$$\operatorname{Re} F(H) = -\frac{1}{8} - \frac{1-\rho^2}{2} u^2 - \frac{1}{2} (Y + \rho u)^2 - s_0 X + \frac{1}{2} X^2. \quad (\text{C.6})$$

If X first reaches a small positive level $\varepsilon < 1/2$, the derivative in (C.3) is positive, whereas (C.6) is negative, a contradiction. Thus $X \leq 0$. Reaching zero at a positive time gives the same contradiction, proving strict negativity.

Let $\psi_\epsilon(z) = \sqrt{|z|^2 + \epsilon^2} - \epsilon$. Equation (C.4), together with

$$\operatorname{Re}(\overline{H}F(H)) = -bX - s_0|H|^2 + \frac{1}{2}X|H|^2, \quad \frac{|H|^2}{\sqrt{|H|^2 + \epsilon^2}} \geq \psi_\epsilon(H),$$

gives $D_C^\alpha \psi_\epsilon(H) + s_0 \psi_\epsilon(H) \leq b$. Compare with the zero-initial-value linear scalar equation and let $\epsilon \downarrow 0$ to obtain (C.5). The scalar solution is verified directly by the Mittag-Leffler series. If finite-time blow-up occurred, (C.5) would bound the trajectory, $F(H)$, and a uniform Hölder constant. The history integral would have a finite limit at that endpoint, and local contraction with the previously accumulated history as forcing would extend the solution, a contradiction. The Caputo history is not restarted. Successive Volterra contractions give uniqueness. ■

Theorem C.2 (dissipative residual bound). Suppose $s_0 > 0$, $\widehat{H}(0) = 0$, and $\widehat{H} \in AC$, with local Lipschitz regularity at positive times. Assume, for every $x \in (0, X]$, that

$$\operatorname{Re} \widehat{H} \leq \epsilon_R < 2s_0, \quad |D_C^\alpha \widehat{H} - F(\widehat{H})| \leq \delta,$$

Set $\sigma = s_0 - \epsilon_R/2 > 0$. Then

$$|H - \widehat{H}| \leq \frac{\delta}{\sigma} [1 - E_\alpha(-\sigma x^\alpha)] \leq \delta/\sigma. \quad (\text{C.7})$$

In particular, for a trajectory in the left half-plane, one may take $\epsilon_R = 0$, $\sigma = s_0$. No smallness assumption on δ or u is required.

Proof. The error $e = H - \widehat{H}$ satisfies

$$D_C^\alpha e = \left(d + \frac{H + \widehat{H}}{2}\right) e - r, \quad e(0) = 0, \quad \operatorname{Re} \left(d + \frac{H + \widehat{H}}{2}\right) \leq -\sigma.$$

Apply (C.4) to $\psi_\epsilon(e)$ to obtain $D_C^\alpha \psi_\epsilon(e) + \sigma \psi_\epsilon(e) \leq |r| \leq \delta$. Comparison with the linear scalar solution, followed by $\epsilon \downarrow 0$, proves the claim. Convex regularisation includes points of zero error and applies to the complex error viewed as a vector in the real plane. ■

If $\widehat{H}$ corresponds to the physical-time trajectory $\widehat{Z}$ and $|r_t|/\nu \leq \delta_F$ has been certified, the same result gives

$$\sup_{t \leq T} |Z - \widehat{Z}| \leq \delta_F/s_0 \quad \text{if } \operatorname{Re} \widehat{Z} \leq 0. \quad (\text{C.8})$$

In the numerical example, $s_0 = 1489/4000$. This dissipation rate controls the complex error modulus; Appendix D.1 supplies the independent residual δ_F . The estimate applies the established convexity tool to the specific error propagation considered here.

The scalar comparison mechanism is established fractional-calculus machinery. Li and Liu [LiLiu2018, Proposition 3.11(ii)] give convexity after regularization and passage to a distributional limit. Their Proposition 4.12 concerns specified vector gradient and Hamiltonian flows. Kopteva [Kopteva2021v2, Lemma 2.8 and Theorem 2.2] uses norm convexity and a positive fractional inverse under positive-time Lipschitz hypotheses. The following elementary bridge proves the vector AC version consumed here directly, including the terminal-time conclusion. It is an adaptation of those tools, not a new general Caputo comparison theorem.

Regularity lemma (AC convexity and the continuous endpoint). Let $0 < \alpha < 1$, $T > 0$, $v \in AC([0, T]; \mathbb{R}^d)$, and let $\Phi \in C^1(\mathbb{R}^d)$ be convex. With $D_C^\alpha v = g_{1-\alpha} * v'$,

$$D_C^\alpha \Phi(v) \leq \nabla \Phi(v) \cdot D_C^\alpha v \quad \text{a.e. on } (0, T). \quad (\text{C.9})$$

Let $u \in AC[0, T]$, $u(0) = 0$, $\lambda \geq 0$, and $R \in L^\infty(0, T)$ be nonnegative. If $D_C^\alpha u + \lambda u \leq R$ almost everywhere, then

$$u(t) \leq (k_\lambda * R)(t), \quad 0 \leq t \leq T, \quad k_\lambda(t) = t^{\alpha-1} E_{\alpha, \alpha}(-\lambda t^\alpha). \quad (\text{C.10})$$

Proof. Choose smooth $f_n \rightarrow v'$ in $L^1(0, T)$, and set $v_n(t) = v(0) + \int_0^t f_n(s) ds$. Then $v_n(0) = v(0)$, $v_n \rightarrow v$ uniformly, and $v'_n \rightarrow v'$ in L^1 . For a smooth path, integration by parts expresses the difference between the two sides of (C.9) as

$$\frac{1}{\Gamma(1-\alpha)} \left[\frac{B_\Phi(v_n(0), v_n(t))}{t^\alpha} + \alpha \int_0^t \frac{B_\Phi(v_n(s), v_n(t))}{(t-s)^{1+\alpha}} ds \right] \geq 0, \quad (\text{C.11})$$

where $B_\Phi(a, b) = \Phi(a) - \Phi(b) - \nabla \Phi(b) \cdot (a - b) \geq 0$. Smooth paths are locally Lipschitz. A gradient bound M on the compact path range and a local path Lipschitz constant L give $B_\Phi(v_n(s), v_n(t)) \leq 2ML|t - s|$. The endpoint kernel $(t - s)^{-\alpha}$ is therefore integrable. The supporting-hyperplane inequality may first be integrated with history truncated at $t - \delta$, and the classical Caputo integrals converge as $\delta \downarrow 0$.

All path ranges lie in one compact set. Continuity of $\nabla \Phi$ and the ordinary AC composition rule give

$$\|\Phi(v_n)' - \Phi(v)'\|_1 \rightarrow 0, \quad \|D_C^\alpha(v_n - v)\|_1 \leq \frac{T^{1-\alpha}}{\Gamma(2-\alpha)} \|v'_n - v'\|_1 \rightarrow 0. \quad (\text{C.12})$$

The same convolution estimate applies to $\Phi(v_n) - \Phi(v)$. Moreover, the products $\nabla \Phi(v_n) \cdot D_C^\alpha v_n$ converge in L^1 to the corresponding product for v . Passing to an almost-everywhere convergent subsequence proves (C.9). This uses the ordinary

composition rule only for the first derivative inside AC ; no Caputo chain rule is asserted.

For (C.10), put $f = D_C^\alpha u + \lambda u \in L^1(0, T)$. The zero initial value gives $u + \lambda g_\alpha * u = g_\alpha * f$. The locally convergent resolvent series, whose n -th term has $L^1(0, T)$ norm bounded by $\lambda^{n-1} T^{n\alpha} / \Gamma(1 + n\alpha)$, gives $u = k_\lambda * f$ almost everywhere. Complete monotonicity of $E_{\alpha, \alpha}(-x)$ [SimonCM2015] supplies $k_\lambda \geq 0$, so $f \leq R$ implies $u \leq k_\lambda * R$ almost everywhere. The convolution on the right is continuous: extend k_λ and R by zero, and use L^1 translation continuity of the kernel with the L^∞ bound on R . Its value at zero is zero. Since u is continuous, an almost-everywhere inequality extends to every point of $[0, T]$, including maturity. ■

C.2. Proof of Theorem 3.1

Proof. For $e = Z - \widehat{Z}$, the divided difference gives $D_C^\alpha e = \nu[d + (Z + \widehat{Z})/2]e - r$, with coefficient real part at most $-\lambda$. Apply the regularity lemma to $v_\varepsilon = (|e|^2 + \varepsilon^2)^{1/2} - \varepsilon$. Since $\nabla v_\varepsilon \cdot e = |e|^2 / (|e|^2 + \varepsilon^2)^{1/2} \geq v_\varepsilon$ and $|\nabla v_\varepsilon| \leq 1$,

$$D_C^\alpha v_\varepsilon + \lambda v_\varepsilon \leq R \quad \text{a.e.} \quad (\text{C.13})$$

Equation (C.10) and the limit $\varepsilon \downarrow 0$ yield the first state inequality at every time. Absolute Fubini gives $I^{1-\alpha} \xi = V_0 g_{2-\alpha} + g_{2-\alpha} * \xi'$, whose derivative is (3.2), and $g_\alpha * q_\alpha = \xi$. In particular, $q_\alpha \in L^1$ and $\xi \geq 0$. Writing $A_\alpha = I^{1-\alpha} \xi$, another absolute Fubini step followed by AC integration by parts gives

$$L_T - \widehat{L}_T = \nu^{-1} \int_0^T A_\alpha(T-t) e'(t) dt = \nu^{-1} (q_\alpha * e)(T). \quad (\text{C.14})$$

The first Fubini integral is bounded by $\|\xi\|_\infty T^{1-\alpha} \|e'\|_1 / \Gamma(2-\alpha)$; the boundary terms vanish because $A_\alpha(0) = e(0) = 0$. Positivity now proves the first exponent estimate. Finally, $g_\alpha - k_\lambda = \lambda g_\alpha * k_\lambda \geq 0$, whence

$$0 \leq q_\alpha * k_\lambda = \xi - \lambda \xi * k_\lambda \leq \xi. \quad (\text{C.15})$$

Associativity and nonnegative integration yield the remaining conclusions. The physical residual is R , so its factor ν^{-1} remains until $R/\nu \leq \delta_F$ is used. ■

The proof also covers $\lambda = 0$: the resolvent is g_α , every convolution remains integrable, and no division by σ occurs.

C.3. Curve positivity and explicit constant-curve bounds

The curve condition in Theorem 3.1 is the nonnegativity of $q_\alpha = (I^{1-\alpha} \xi)'$, rather than monotonicity of ξ . The state and reference still belong to $AC[0, T]$, have the same zero initial value, and satisfy the full-time residual and dissipation hypotheses in (3.1). The physical residual remains R , the normalized residual is R/ν , and physical damping is $\lambda = \nu\sigma > 0$. Both exponents retain the D-type definitions in Section 3. These assumptions are required for each application; the condition on the pricing kernel does not prove stochastic-model existence or extend the verified Riccati regularity domain.

For a real curve $\xi \in AC[0, T]$ with $\xi(0) = V_0 \geq 0$, the precise sufficient condition is

$$q_\alpha = V_0 g_{1-\alpha} + g_{1-\alpha} * \xi' \geq 0 \quad \text{a.e.}, \quad g_\beta(t) = t^{\beta-1}/\Gamma(\beta). \quad (\text{C.16})$$

No sign is imposed on ξ' . Indeed, $A_\alpha = I^{1-\alpha}\xi = V_0 g_{2-\alpha} + g_{2-\alpha} * \xi'$ is absolutely continuous, $A_\alpha(0) = 0$, and

$$\|q_\alpha\|_1 \leq \frac{T^{1-\alpha}}{\Gamma(2-\alpha)}(V_0 + \|\xi'\|_1), \quad g_\alpha * q_\alpha = \xi. \quad (\text{C.17})$$

Absolute Fubini proves the second identity even for signed ξ' , using $g_\alpha * g_{1-\alpha} = g_1$. The initial term $V_0 g_{1-\alpha}$ cannot be dropped. Condition (C.16) implies $\xi \geq 0$. Hence the entire proof of Theorem 3.1 continues to hold: integration by parts gives $L_T - \widehat{L}_T = \nu^{-1}(q_\alpha * e)(T)$, and the positive resolvent identity gives

$$0 \leq q_\alpha * k_\lambda = \xi - \lambda \xi * k_\lambda \leq \xi. \quad (\text{C.18})$$

The Fubini step for the exponent is bounded by $\|\xi\|_\infty T^{1-\alpha} \|e'\|_1 / \Gamma(2-\alpha)$; its boundary terms vanish because $e(0) = A_\alpha(0) = 0$. The AC Caputo convexity and the positive zero-initial-value inverse remain the prior tools [LiLiu2018, Kopteva2021v2]. The finite-history convolution, cell-weight bound (3.5), and outward node-radius construction therefore remain valid under (C.16). Cells still carry the complete Caputo history.

This is a strict relaxation of a sufficient curve condition, not a necessary condition for every conceivable certificate. If q_α changes sign, the safe general estimate is $\nu^{-1}(|q_\alpha| * k_\lambda * R)(T)$; the positive collapse to $\xi * R$ cannot be used without further proof. Fractional forward-variance representations and their model-admissibility restrictions already occur in [ElEuchRosenbaum2017v1, Proposition 3.1 and Corollary 3.3]. Our analytical condition does not replace those restrictions.

Corollary C.3 (a decreasing analytical curve). Let $V_0 > 0$, $c > 0$, and $\xi(t) = V_0(1 - ct)$. Then

$$q_\alpha(t) = \frac{V_0 t^{-\alpha}}{\Gamma(1-\alpha)} \left(1 - \frac{ct}{1-\alpha}\right). \quad (\text{C.19})$$

Thus (C.16) holds on $(0, T)$ if and only if $cT \leq 1 - \alpha$, including equality. In that case $\xi(T) \geq \alpha V_0 > 0$ despite $\xi' < 0$.

Proof. Convolution of the constant derivative $-V_0 c$ with $g_{1-\alpha}$ gives $-V_0 c t^{1-\alpha} / \Gamma(2-\alpha)$. Adding the initial term and using $\Gamma(2-\alpha) = (1-\alpha)\Gamma(1-\alpha)$ proves (C.19). Its bracket is nonnegative exactly under the stated condition. ■

This example is analytical and has not been used as a new financial experiment or asserted to be a realized rough-Heston variance curve. Positivity of the curve alone is insufficient for (C.16): $1 - \alpha < cT < 1$ leaves $\xi > 0$ but gives a negative kernel near maturity.

Corollary C.4 (explicit constant-curve propagation). Under Theorem 3.1's state hypotheses with $\sigma > 0$, let $\xi \equiv V_0 \geq 0$. Then

$$q_\alpha * k_\lambda = V_0 E_\alpha(-\lambda t^\alpha), \quad |L_T - \widehat{L}_T| \leq \frac{V_0}{\nu} \int_0^T E_\alpha(-\lambda(T-s)^\alpha) R(s) ds. \quad (\text{C.20})$$

If $R \leq \nu\delta_F$, a closed-form upper bound is

$$\eta_{\text{ML}} = V_0\delta_F TE_{\alpha,2}(-\nu\sigma T^\alpha) \leq \min \left\{ V_0\delta_F T, \frac{V_0\delta_F T^{1-\alpha}}{\nu\sigma\Gamma(2-\alpha)} \right\}. \quad (\text{C.21})$$

Proof. Termwise convolution in the absolutely locally integrable Mittag-Leffler series gives $g_{1-\alpha} * k_\lambda = E_\alpha(-\lambda t^\alpha)$; termwise integration gives $\int_0^T E_\alpha(-\lambda t^\alpha) dt = TE_{\alpha,2}(-\lambda T^\alpha)$. Equation (C.18) gives $0 \leq E_\alpha \leq 1$, proving the first comparison. Also $\int_0^t k_\lambda = (1 - E_\alpha(-\lambda t^\alpha))/\lambda \leq 1/\lambda$. The state envelope $|e| \leq \delta_F/\sigma$, followed by the positive q_α integral, proves the second comparison. ■

The explicit bound retains finite history and physical scaling. It may be intersected with earlier independently valid exponent bounds only when all bound the same D-type target and reference. A substitution-based reference requires its separately certified residual conversion. This constant-curve corollary is analytical; no separate constant-curve model experiment is claimed. Section 7.6 implements the resolvent kernel for the fixed nonconstant curve.

Proposition C.5 (an error radius without an assumed approximate half-plane). Assume $s_0 > 0$, $0 \leq \delta_F < s_0^2/2$, $D_C^\alpha Z = \nu F(Z)$, $\Re Z \leq 0$, and $|D_C^\alpha \widehat{Z} - \nu F(\widehat{Z})| \leq \nu\delta_F$ almost everywhere. The exact and approximate trajectories are AC, locally Lipschitz at positive times, and have the same zero initial value. Then

$$|Z - \widehat{Z}| \leq E_\delta = s_0 - \sqrt{s_0^2 - 2\delta_F} = \frac{2\delta_F}{s_0 + \sqrt{s_0^2 - 2\delta_F}}. \quad (\text{C.22})$$

To prove this, rewrite the error terms as $(d + Z)e - e^2/2$. The exact-solution half-plane gives $D_t^\alpha |e| \leq \nu(-s_0|e| + |e|^2/2 + \delta_F)$, interpreted through the same convex regularisation of the modulus. Choose any $E_\delta < \ell < s_0 + \sqrt{s_0^2 - 2\delta_F}$. The right-hand side is strictly negative at $|e| = \ell$; a first hitting time of ℓ contradicts (C.3). Letting $\ell \downarrow E_\delta$ proves the result. This is a sufficient small-residual condition; violating it does not establish an actual error or pole. The completed node trajectories in this example have a certified half-plane property and use (C.8), which does not impose the threshold in (C.22).

C.4. Complementary global exponent bound

For (2.4), $\xi_* \in AC$, $\xi'_* \geq 0$, and $V_0 \leq \xi_* \leq \theta$. Positivity of the scalar resolvent follows from the negative-minimum principle in (C.3), or from the completely monotone representation of Simon [SimonCM2015]. Define

$$A_\alpha(t) = (I^{1-\alpha}\xi_*)(t), \quad q_\alpha(t) = A'_\alpha(t) = \frac{V_0 t^{-\alpha}}{\Gamma(1-\alpha)} + I^{1-\alpha}\xi'_*(t) \geq 0. \quad (\text{C.23})$$

Both initial exponents $-\alpha, \alpha_0 - \alpha$ exceed -1 , so $q_\alpha \in L^1(0, T)$. Explicitly,

$$q_\alpha(t) = \frac{V_0 t^{-\alpha}}{\Gamma(1-\alpha)} + (\theta - V_0)\lambda_\xi t^{\alpha_0 - \alpha} E_{\alpha_0, 1 + \alpha_0 - \alpha}(-\lambda_\xi t^{\alpha_0}), \quad (\text{C.24})$$

$$A_\alpha(t) = \frac{\theta t^{1-\alpha}}{\Gamma(2-\alpha)} + (V_0 - \theta)t^{1-\alpha}E_{\alpha_0, 2-\alpha}(-\lambda_\xi t^{\alpha_0}), \quad A_\alpha(0) = 0. \quad (\text{C.25})$$

where $E_{a,b}(z) = \sum_{n=0}^{\infty} z^n / \Gamma(an + b)$.

Theorem C.6. For a zero-initial-value AC trajectory $Z, \widehat{Z}$, use the derivative-based exponent

$$L_T = \nu^{-1} \int_0^T \xi_*(T-t) D_t^\alpha Z(t) dt, \quad \bar{L}_T = \nu^{-1} \int_0^T \xi_*(T-t) D_t^\alpha \widehat{Z}(t) dt,$$

Then

$$L_T - \bar{L}_T = \nu^{-1} \int_0^T q_\alpha(T-t)(Z - \widehat{Z})(t) dt. \quad (\text{C.26})$$

In particular, if $\sup |Z - \widehat{Z}| \leq E$, then

$$|L_T - \bar{L}_T| \leq \eta = EA_\alpha(T)/\nu \leq \frac{E\theta T^{1-\alpha}}{\nu\Gamma(2-\alpha)}. \quad (\text{C.27})$$

Proof. Since $D_t^\alpha v = I^{1-\alpha}v'$, absolute Fubini changes the exponent integral into $\nu^{-1} \int_0^T A_\alpha(T-t)v'(t) dt$. Integration by parts, using $A_\alpha(0) = v(0) = 0$, yields the positive-kernel representation. Absolute Fubini is justified by $v' \in L^1$ and the continuous, finite curve. Then $q_\alpha \geq 0$ and $\int q_\alpha = A_\alpha(T)$ give the estimate. ■

If $\widehat{Z} = I^\alpha \bar{G}$, then $\bar{L} = \nu^{-1} \int \xi_* \bar{G}$ is exactly this derivative-based exponent, so the residual integral is not added again. The substitution-based exponent formed from $\int \xi_* F(\widehat{Z})$ differs by $\nu^{-1} \int \xi_* r_t$ and requires a separate conversion. All numerical exponent quantities used here are derivative-based.

Using $\text{Re } L_T \leq 0$ and the integral identity for the exponential, whenever (C.27) holds,

$$|e^{L_T} - e^{\bar{L}_T}| \leq \min\{1, e^{\text{Re } \bar{L}_T}\}(e^\eta - 1). \quad (\text{C.28})$$

The first estimate is based at the exact exponent; the second is based at the approximate exponent. Their minimum is a valid bound. The state error may increase with frequency, while decay of the approximate transform under the pricing weight can compensate for part of that increase.

If a model-transform envelope $B(u)$ and an approximate-modulus bound $\widehat{B}(u)$ are also available, then

$$|e^{L_T} - e^{\bar{L}_T}| \leq \min\{B + \widehat{B}, \eta(B + \widehat{B})/2\}. \quad (\text{C.29})$$

The first bound is the triangle inequality. The second follows from $(L_T - \bar{L}_T) \int_0^1 e^{(1-\tau)\bar{L}_T + \tau L_T} d\tau$ and the convexity bound $e^{(1-\tau)a + \tau b} \leq (1-\tau)e^a + \tau e^b$. It can be combined with (C.28) by taking the minimum, or compared with the error bound $B(u)$ obtained by explicitly setting the node approximation to zero. The calculation uses the smallest of the applicable bounds.

C.5. Certified dissipative-kernel weights

Retain the full assumptions and AC comparison proof of Theorem 3.1, but allow $\sigma \geq 0$. The regularized error modulus satisfies $D_C^\alpha v_\varepsilon + \nu \sigma v_\varepsilon \leq R$ almost everywhere. The same positive inverse applies when $\lambda = \nu \sigma = 0$: $k_0 = g_\alpha$. Passing to the continuous endpoint and then composing with the pricing functional gives (3.6)-(3.7). In particular the zero-damping assertion introduces no $1/\sigma$. The initial contribution $V_0 g_{1-\alpha}$ in q_α and the factor ν^{-1} are unchanged.

Proposition (a rigorous cumulative-weight series). Suppose $0 < \alpha, A_0 < 1$, $0 \leq V_0 \leq \theta$, and the fixed curve is $\xi(t) = \theta + (V_0 - \theta)E_{A_0}(-\lambda_\xi t^{A_0})$, with $\lambda_\xi \geq 0$. Retain $q_\alpha \geq 0$. For $\lambda \geq 0$ and $\lambda T^\alpha < 1$, (3.8) holds on $[0, T]$. If it is truncated before index M , the entire omitted outer series satisfies

$$\left| W_\lambda(t) - \sum_{n=0}^{M-1} (-\lambda)^n W_n(t) \right| \leq \frac{\theta t (\lambda t^\alpha)^M}{1 - \lambda t^\alpha}, \quad W_n(t) = \int_0^t (\xi * g_{n\alpha})(s) ds. \quad (\text{C.30})$$

Here g_0 denotes the identity convolution, so $W_0(t) = \int_0^t \xi$.

Proof. The existing resolvent identity $k_\lambda + \lambda g_\alpha * k_\lambda = g_\alpha$, together with $q_\alpha * g_\alpha = \xi$, gives $K_\lambda + \lambda g_\alpha * K_\lambda = \xi$. The Neumann series has terms $(-\lambda)^n (\xi * g_{n\alpha})$. Since $0 \leq \xi \leq \theta$,

$$0 \leq W_n(t) \leq \frac{\theta t^{1+n\alpha}}{\Gamma(2+n\alpha)} \leq \theta t (t^\alpha)^n. \quad (\text{C.31})$$

Log convexity and $\Gamma(1) = \Gamma(2) = 1$ imply that Γ is increasing and at least one on $[2, \infty)$. Every outer denominator argument is $2 + n\alpha \geq 2$. Thus the sum of the $L^1(0, T)$ norms of the absolute terms is bounded by the convergent geometric series in (C.31). This justifies convolution, integration and interchange, proves the Volterra identity and its unique resolvent solution, and gives (C.30) with the first omitted index exactly $n = M$. In particular twelve retained terms omit $n \geq 12$, rather than $n \geq 13$.

For uniqueness, the difference $h \in L^1(0, T)$ of two solutions satisfies $h = -\lambda g_\alpha * h$. Iterating gives $\|h\|_1 \leq \lambda^n T^{n\alpha} \|h\|_1 / \Gamma(1 + n\alpha)$. For sufficiently large n , the Gamma argument is at least two, and the right-hand factor is at most $(\lambda T^\alpha)^n$, which tends to zero. Thus $h = 0$. This argument does not assume an unjustified one-step contraction involving $\Gamma(1 + \alpha)$.

Expanding the curve Mittag-Leffler function, and using the beta convolution identity, gives

$$W_n(t) = t^{1+n\alpha} \left[\frac{\theta}{\Gamma(2+n\alpha)} + (V_0 - \theta) \sum_{m=0}^{\infty} \frac{(-\lambda_\xi t^{A_0})^m}{\Gamma(2+n\alpha+mA_0)} \right]. \quad (\text{C.32})$$

When $w = \lambda_\xi t^{A_0} < 1$, every inner denominator also has argument at least two, and its tail before index L is bounded by

$$\left| \sum_{m=L}^{\infty} \frac{(-w)^m}{\Gamma(2+n\alpha+mA_0)} \right| \leq \frac{w^L}{1-w}. \quad (\text{C.33})$$

The tails are absolute enclosures; no unproved alternating-series monotonicity is needed. The actual curve and damping satisfy the strict interval checks $w < 1$ and $\lambda T^\alpha < 1$. Signed coefficients, including $V_0 - \theta$, are multiplied by outward intervals before addition. Every power, Gamma value and exponential is enclosed by the identified dyadic primitives. ■

For a residual envelope $R \leq R_j$ on the complete cell $[a_j, b_j]$, evaluate strict cumulative intervals $[W^-(t), W^+(t)]$. Its exact weight lies in

$$[W^-(T - a_j) - W^+(T - b_j), W^+(T - a_j) - W^-(T - b_j)]. \quad (\text{C.34})$$

Intersecting with nonnegativity and the independently proved upper curve weight is valid because $0 \leq K_\lambda \leq \xi$. The resulting upper endpoint ω_j^+ gives

$$\eta_{\text{res}} \leq \nu^{-1} \sum_j R_j \omega_j^+. \quad (\text{C.35})$$

Every bin envelope is the maximum over all intersecting closed source cells, including endpoint intersections. Source cells still start at zero and cover the full interval; no Caputo history is restarted.

The producer evaluates W_n through the existing strict power-field moment divided by $\Gamma(1 + n\alpha)$, retaining twelve outer terms. The separate reader evaluates (C.32) directly, retaining fourteen outer terms and sixty-four inner terms, without importing the producer's cumulative function or the power-field-moment function. It verifies that the claimed residual-functional enclosure contains its tighter independent enclosure, and checks all transform-radius minima and full prices. The Gamma, logarithm, exponential and dyadic coefficient primitives remain shared dependencies; the upstream residual proof is inherited rather than independently rederived.

Appendix D. Continuous certification and reliable refinement

D.1. Reference trajectory, startup and arithmetic

For each candidate $\alpha = \beta \in \{.52, .6, .9\}$ and each $u = n/8$, $n = 0, \dots, 512$, we store

$$\bar{G}(t) = \nu c_0 + A_1 t^\beta + A_2 t^{2\beta} + \bar{L}(t), \quad c_0 = -(u^2 + 1/4)/2, \quad \hat{Z} = I^\alpha \bar{G}, \quad T = 1/2. \quad (\text{D.1})$$

Here $\bar{L}$ is the continuous piecewise-linear interpolation in physical time at nodes (t_j, L_j) , with $L_0 = 0$. Nodes and complex coefficients are interpreted as the exact dyadic rationals represented by their binary64 values. The following independent residual controls the continuous-time error of this reference trajectory:

$$r_t = \bar{G} - \nu F(\hat{Z}), \quad \delta_F = \nu^{-1} \sup_{0 \leq t \leq T} |r_t|. \quad (\text{D.2})$$

Equation (D.1) defines an entire absolutely continuous reference trajectory. Its independent residual gives the model-price intervals, which are then used to bound the output error of the selected [3/3] implementation.

Write

$$\widehat{Z} = H_0 + J, \quad J = I^\alpha \bar{L}, \quad H_0 = B_1 t^\alpha + B_2 t^{\alpha+\beta} + B_3 t^{\alpha+2\beta},$$

$$B_1 = \frac{\nu c_0}{\Gamma(1+\alpha)}, \quad B_2 = \frac{A_1 \Gamma(1+\beta)}{\Gamma(1+\alpha+\beta)}, \quad B_3 = \frac{A_2 \Gamma(1+2\beta)}{\Gamma(1+\alpha+2\beta)}. \quad (\text{D.3})$$

The initial interval is $J = L_1 t^{\alpha+1}/[t_1 \Gamma(2+\alpha)]$. Substitute all generalised powers into (D.2), first cancel νc_0 exactly, combine equal powers, and then take $\sum_p |r_p| t_1^p$. Since every $p > 0$, this bounds the full initial closed interval. The generalised-power expansion retains the fractional initial behaviour.

On a source interval $[a, b] \subset [0, q]$, set $\tau = q - a, h = b - a$. The integral weights for the linear hat functions are

$$I_0 = \{\tau^\alpha - (\tau - h)^\alpha\}/\alpha, \quad I_1 = \{\tau^{\alpha+1} - (\tau - h)^{\alpha+1}\}/(\alpha + 1),$$

$$w_R = (\tau I_0 - I_1)/(h \Gamma(\alpha)), \quad w_L = I_0/\Gamma(\alpha) - w_R. \quad (\text{D.4})$$

These weights are nonnegative. For a truncated interval, recover the endpoints by affine interpolation within the original interval. On the final interval, $b = q$ can be evaluated directly using $w_R = h^\alpha/[\alpha(\alpha+1)\Gamma(\alpha)]$, $w_L = \alpha w_R$. For $h/\tau < .01$, retain eight positive-series terms, using respectively

$$w_R = \frac{h\tau^{\alpha-1}}{\Gamma(\alpha)} \sum_{k \geq 0} \frac{(1-\alpha)_k (h/\tau)^k}{k!(k+2)}, \quad w_L = \frac{h\tau^{\alpha-1}}{\Gamma(\alpha)} \sum_{k \geq 0} \frac{(1-\alpha)_k (h/\tau)^k}{k!(k+1)(k+2)}. \quad (\text{D.5})$$

The unscaled tails are each bounded by $z^8/(1-z)$ because $0 < (1-\alpha)_k/k! \leq 1$. This reduces interval inflation from subtracting nearly equal endpoint quantities while retaining rigorous history integration.

Subdivide each subsequent original interval twice into four closed subintervals $[a, b]$, with recorded dyadic centres m . Given a derivative bound on the full subinterval,

$$\sup_{[a,b]} |r_t| \leq |r_t(m)| + \max(m-a, b-m) \sup_{[a,b]} |r'_t|. \quad (\text{D.6})$$

The ordinary residual derivative may be discontinuous at original nodes. The derivative bounds below hold almost everywhere; absolute continuity of the residual and the fundamental theorem of calculus still imply (D.6). Each of the following three identities provides a derivative bound, and we take their minimum:

$$r'_t = \bar{G}' - \nu(d + \widehat{Z})\widehat{Z}',$$

$$P_r = \nu c_0 + A_1 t^\beta + A_2 t^{2\beta} - \nu F(H_0), \quad r_t = P_r + \bar{L} - \nu(d + H_0)J - \nu J^2/2,$$

$$r'_t = P'_r + \bar{L}' - \nu[(d + H_0 + J)J' + H'_0 J], \quad \widehat{Z}' = \frac{\nu c_0 t^{\alpha-1}}{\Gamma(\alpha)} + I^\alpha \bar{G}'. \quad (\text{D.7})$$

The constant-kernel weight of a past source interval decreases with the integration endpoint, whereas the weight of the current source interval increases while that endpoint lies within it. Endpoint bounds therefore enclose both types of weight. In the third identity, first combine $\bar{G}'(s) = \beta A_1 s^{\beta-1} + 2\beta A_2 s^{2\beta-1} + \bar{L}'(s)$ on each source

interval and then integrate against the positive kernel, retaining cancellation in the original relation.

The singular terms from the initial source interval are handled directly by fractional-power integrals. For $p > 0, t_1 \leq a \leq q \leq b$, the positive weight satisfies

$$\frac{t_1^p b^{\alpha-1}}{p\Gamma(\alpha)} \leq \frac{1}{\Gamma(\alpha)} \int_0^{t_1} s^{p-1} (q-s)^{\alpha-1} ds \leq \min \left\{ \frac{t_1^p (a-t_1)^{\alpha-1}}{p\Gamma(\alpha)}, \frac{\Gamma(p)}{\Gamma(\alpha+p)} \max_{q \in [a,b]} q^{\alpha+p-1} \right\}. \quad (\text{D.8})$$

When $a = t_1$, discard the first upper bound. The second follows from the full Euler Beta integral and is finite. Kernel monotonicity at the endpoints gives the lower bound and the first upper bound. Every $p, \alpha > 0$, so the hypotheses of the classical Beta formula hold.

Using the same $\widehat{Z}'$ enclosure and rigorous point values, certify an upper bound for $\text{Re } \widehat{Z}$ on every subsequent subinterval. On the initial interval, factor out t^α , leaving a finite bracket; use its negative constant term and the endpoint contributions of the remaining positive parts. Bound (C.8) is used only when the upper real-part bound is nonpositive on every closed interval. This condition therefore covers the entire closed time interval.

The structural certificate and special functions use outward-rounded intervals with integer endpoints divided by 2^{100} . Batched residual weights use IEEE binary64; each basic operation is enlarged outward to adjacent representable numbers. After converting an exact rational Gamma enclosure, endpoint inclusion is checked with Fraction. The logarithm uses exact binary scaling and a twenty-two-term atanh series with $z \in [0, 1/3]$, whose tail is $2z^{45}/[45(1-z^2)]$. After scaling, the exponential has $|r| < 1$ and a degree-24 Taylor tail bounded by $3/25!$. These explicit remainders define the intervals for logarithms, exponentials, and noninteger powers.

For matrix multiplication with rigorous weight centres W_c , radii W_r , and recorded dyadic values X , include

$$(\gamma_{2n} \|W_c\|_{1,\text{row}} + \|W_r\|_{1,\text{row}}) \|X\|_{\infty,\text{column}}, \quad \gamma_{2n} = \frac{2n 2^{-53}}{1 - 2n 2^{-53}}, \quad (\text{D.9})$$

and an additional underflow-error allowance. Induction on products of basic rounding factors gives this estimate and permits different summation association orders. Norm summations are also rounded outward. The calculation assumes round-to-nearest IEEE basic operations, correctly rounded square roots, gradual underflow, and the absence of overflow or NaNs. The supplementary material records source and computation versions. The interval-inclusion conclusions in this section are conditional on these arithmetic assumptions and the stated remainder bounds.

For the fixed curve, define

$$J_0(z) = \theta z + (V_0 - \theta) z E_{\alpha_0,2}(-\lambda_\xi z^{\alpha_0}), \quad J_1(z) = \theta z^2/2 + (V_0 - \theta) z^2 (E_{\alpha_0,2} - E_{\alpha_0,3})(-\lambda_\xi z^{\alpha_0}). \quad (\text{D.10})$$

These quantities are respectively $\int_0^z \xi_*(s) ds, \int_0^z s \xi_*(s) ds$; termwise integration verifies the second formula. If a linear-trajectory interval is $[a, b]$ with $A = T - b, B = T - a$, its rigorous nonnegative exponent weights are

$$w_l = \frac{J_1(B) - J_1(A) - A[J_0(B) - J_0(A)]}{b - a}, \quad w_r = \frac{B[J_0(B) - J_0(A)] - J_1(B) + J_1(A)}{b - a}. \quad (\text{D.11})$$

The curve moment for the power component t^β is

$$\int_0^T \xi_*(T - t)t^\beta dt = \frac{\theta T^{\beta+1}}{\beta + 1} + (V_0 - \theta)\Gamma(\beta + 1)T^{\beta+1}E_{\alpha_0, \beta+2}(-\lambda_\xi T^{\alpha_0}). \quad (\text{D.12})$$

This follows directly from the series and Beta integration. Together with (D.1), it makes $\bar{L}$ a rigorous finite sum. Trigonometric phases and complex exponentials also use finite Taylor expansions with explicit tails. On the present horizon, if $z = \lambda_\xi T^{\alpha_0} < 1$, a Mittag-Leffler series tail may be bounded by $3z^{64}/(1 - z)$: all Gamma arguments exceed one, and Euler's integral gives $\Gamma(\gamma) \geq e^{-1} > 1/3$. Rigorous integration encloses the approximate exponent, while (C.27) controls the difference between the model and approximate exponents.

D.2. Strip, exact high-frequency tail and reference field

Proof. Jensen's inequality gives, within the strip, $|\phi_T(z - i/2)| \leq \mathbb{E}M_T^{1/2 - \text{Im}z} \leq 1$. On compact substrips, $x^q |\log x|^j \leq C(1 + x)$, so the transform is analytic. The boundaries $z = r \pm ia_*$ satisfy

$$|z^2 + 1/4| \geq r^2 + (1/2 - a_*)^2, \quad \int_{\mathbb{R}} |g(r \pm ia_*)| dr \leq M_* = \frac{\pi e^{a_*|k|}}{1/2 - a_*}.$$

Shift the real-axis Fourier integration contour to the upper and lower boundaries. The vertical integrals vanish by quadratic decay, giving $|\tilde{g}(\omega)| \leq M_* e^{-a_*|\omega|}$. The absolutely and uniformly convergent periodisation $\sum_{j \in \mathbb{Z}} g(x + jh_*)$ has Fourier coefficients $h_*^{-1} \tilde{g}(2\pi n/h_*)$. Summing at zero and subtracting $n = 0$ gives a two-sided error of at most $2M_*/(e^{2\pi a_*/h_*} - 1)$. Conjugate symmetry halves both the two-sided integral and its sum. Multiplication by $\sqrt{m}/\pi$ gives (4.2). ■

This is a specific application of the analytic-strip trapezoidal theory of Trefethen and Weideman [TrefethenWeideman2014]. Each pricing node uses its own continuous-time residual bound; the analytic strip controls the quadrature error between nodes.

Fix $\kappa = 0, -1 < \rho < 0$. Set $X = -\text{Re} H \geq 0$. Equation (C.6) gives

$$D_x^\alpha X \geq \beta_u - s_0 X - X^2/2, \quad \beta_u = (1 - \rho^2)u^2/2 + 1/8. \quad (\text{D.13})$$

Let w be the scalar zero-initial-value solution satisfying equality. Comparison gives

$$X \geq w, \quad 0 \leq w \leq R_u, \quad R_u = \sqrt{s_0^2 + 2\beta_u} - s_0, \quad \ell_u = (s_0 + \sqrt{s_0^2 + 2\beta_u})/2.$$

Because the right-hand side satisfies $(R_u - w)(s_0 + (R_u + w)/2) \geq \ell_u(R_u - w)$, linear comparison and the bound of Simon [Simon2014], $E_\alpha(-z) \leq (1 + z/\Gamma(1 + \alpha))^{-1}$, give

$$-\text{Re} Z(t, u) \geq R_u \frac{\ell_u \nu t^\alpha}{\Gamma(1 + \alpha) + \ell_u \nu t^\alpha}. \quad (\text{D.14})$$

The comparison argument uses a positive maximum over the full history.

Take $V > s_0/\sqrt{1-\rho^2}$ and define

$$a_\rho = \sqrt{1-\rho^2}, \quad b_V = a_\rho - s_0/V > 0, \quad f_\alpha(z) = z/(\Gamma(1+\alpha) + z).$$

For $u \geq V$, we have $R_u/u \geq b_V, \ell_u \geq a_\rho u/2$. For any partition of $J \geq 2$, denoted $0 = t_0 < \dots < t_J = T$, the positive kernel in (C.26) and the monotonicity of f_α give

$$c_V = \frac{b_V}{\nu} \sum_{j=0}^{J-1} f_\alpha(a_\rho \nu V t_j^\alpha / 2) [A_\alpha(T - t_j) - A_\alpha(T - t_{j+1})] > 0,$$

$$|\phi_T(u - i/2)| \leq e^{-c_V u} \quad (u \geq V). \quad (\text{D.15})$$

This envelope holds at every continuous high frequency. The time partition constructs a lower sum for the positive-measure integral. The example uses $J = 64$ and rigorous intervals to obtain a positive rational lower bound for c_V . No singular evaluation is performed at $A_\alpha(0) = 0$. The difference between bounds taken from the appropriate sides at distinct endpoints, $[A_{\text{lo}}(T - t_j) - A_{\text{hi}}(T - t_{j+1})]_+$, is a valid lower bound on the mass.

Using the right-endpoint sum of a positive decreasing function, for $V = N_u h_*$ the exact discrete-tail contribution to the normalised price is bounded by

$$\epsilon_{\text{tail}} \leq \frac{\sqrt{m}}{\pi} \frac{e^{-c_V V}}{c_V V^2}. \quad (\text{D.16})$$

The integral tail uses an envelope for the exact solution. Finite nodes assigned the value zero are charged according to the same envelope, so every part of the frequency range has an explicit error contribution.

For each candidate, a separately implemented product-integration procedure generates and fixes a physical-time reference function $\bar{G} = \nu c_0 + A_1 t^\beta + A_2 t^{2\beta} + \bar{L}$, where $\beta = \alpha_j$. Its nodes and coefficients are interpreted as their recorded exact dyadic values. The certified trajectory is $\hat{Z} = I^\alpha \bar{G}$, rather than the floating-point PI nodes themselves. The initial time interval uses the full generalised-power residual expansion. The remaining closed intervals use rigorous convolution-derivative bounds and the mean value theorem, covering all $t \in [0, 1/2]$. At each Fourier node, division of the physical residual by exact ν gives the input to the proved error barrier or certified half-plane linear error estimate.

The recorded exponent is derivative-based: $\bar{L}_T = \nu^{-1} \int \xi_*(T - t) \bar{G}(t) dt$. The positive kernel $q_\alpha = (I^{1-\alpha} \xi_*)' \geq 0$ gives

$$|L_T - \bar{L}_T| \leq E_{\text{state}}(I^{1-\alpha} \xi_*)(T)/\nu \leq E_{\text{state}} \theta T^{1-\alpha}/[\nu \Gamma(2-\alpha)]. \quad (\text{D.17})$$

No residual integral is added a second time. Gamma and Mittag-Leffler curve moments, trajectory integrals, exponents, phases, and sums for the fixed curve use outward-rounded 100-bit rational intervals.

The positive-half-axis price rule has $h = 1/8$ and analytic-strip width $a = 9/20$. At the available nodes $u = 0 : 1/8 : 64$, the recorded trajectories use their actual residual certificates. For nodes above 64 through 128, the approximate transform is explicitly

zero and each error is bounded by the exact-transform envelope. Above 128, the proved infinite discrete-rule tail bound obtained from the proved continuous-integral envelope is included. A comparison bound for the real part of the exact Riccati solution and a 64-cell lower sum over the entire positive-kernel curve give the high-frequency envelope. The analytic-strip trapezoidal error is estimated separately. Passage from the full-axis integral to the positive half-axis, including the half-weight at zero, is handled explicitly. The resulting enclosures cover all twelve model prices in Appendix E and are based on error bounds rather than grid-convergence differences or Richardson diagnostics.

Here E_{state} is any valid certified state radius: the expression E_δ of Proposition C.5 requires $\delta_F < s_0^2/2$; at other nodes the certified half-plane bound gives $E_{\text{state}} = \delta_F/\sigma$. The complete new experiments primarily use the finite-history exponent envelope of Theorem 3.1, and retain this global alternative only when valid and tighter. The nearby fields use the same construction with independently generated coefficients and continuous residual banks.

D.3. Nested certificates and the fixed-menu invariant

Proposition D.1 (safe finite refinement). Fix the output, reference, coefficients, and complete remainder. Replacing a node radius by the smaller of its old radius and a newly proved radius yields a nested joint error set. Consequently every task direction has a nonincreasing certified radius. For a task direction w , choose R_w to bound the remainder support in both directions w and $-w$, and rational U_n to bound the common node coefficient modulus. The symmetric remainder construction used in the numerical examples satisfies this requirement. Let

$$B_0 \geq R_w + \sum_n U_n \varepsilon_n^0, \quad g = B_0 - R_w - \sum_n U_n \varepsilon_n^0 \geq 0. \quad (\text{D.18})$$

An exact update $B \leftarrow B - U_n(\varepsilon_n^{\text{old}} - \varepsilon_n^{\text{new}})$ preserves the invariant $B = g + R_w + \sum_n U_n \varepsilon_n$. A financial budget is returned as certified only when $DF(|w^\top d| + B) \leq \tau$. An initially sufficient bound needs zero actions. If all finitely many admissible updates suffice and the policy visits every unprocessed action, it terminates in at most that many actions. Otherwise it returns the valid unresolved interval; this does not prove that the actual error exceeds the budget.

Proof. Disk inclusion is preserved by finite products, the pricing map, and addition of the same remainder. Support is monotone under inclusion. The invariant follows by subtracting one nonnegative exact decrement while retaining the initial rounding margin. The stopping and finite-exhaustion statements follow from the invariant and the finite action list. ■

Choosing the current largest $U_n \varepsilon_n$ is a valid scheduling policy, not a theorem of minimum cost. If all alternative radii have already been certified and every upgrade is assigned unit cost, sorting the exact decrements $U_n(\varepsilon_n^0 - \varepsilon_n^1)$ gives the minimum number of upgrades needed to satisfy a fixed scalar budget. The exchange argument replaces any selected smaller decrement by an unselected larger decrement without reducing progress. The 82-upgrade result in Appendix E concerns only the specified global-maximum-residual finite-history menu; it does not assert optimality for the separately computed time-local radii or other menus. Precomputing the alternative radii

is real work and must be included in total cost; minimum action count does not establish minimum resource cost or optimal continuous-residual generation.

D.4. Finite-objective and perturbation proofs

Proof. The exact difference $c_i - M_i$ belongs to the closed interval used in (5.14). The minimum of its square is ℓ and the maximum is v , so the weighted sum encloses every J_j . Each $J_j \geq L_j \geq L_*$; a candidate attaining the smallest U_j gives $J_* \leq J_j \leq U_*$, proving (5.15). If $J_j \leq J_* + \varepsilon$, then $L_j \leq J_j \leq U_* + \varepsilon$, proving (5.16). Conversely, $J_j - J_* \leq U_j - L_*$ gives the stated sufficient inner condition. Under (5.17), every competitor satisfies $J_j - J_{j_0} \geq L_j - U_{j_0} \geq g > 0$, proving uniqueness. Condition (5.18) places the model objective of every selected higher-H candidate above $J_* + \varepsilon$ and therefore excludes it. No independence assumption on quote errors across candidates is needed.

If $p^+ < b^-$, then the exact price satisfies $c < B$; if $p^- > a^+$, then $c > A$. A single such row rules out simultaneous consistency. In the opposite direction, the inner-band conditions give $B \leq b^+ \leq c \leq a^- \leq A$. Applying these statements row by row proves the consistency assertions. ■

Corollary (objective perturbation). Suppose another numerical objective $\tilde{J}$ satisfies $\sup_{\Theta_{\text{finite}}} |\tilde{J} - J| \leq \delta_J$, and its selected candidate is ε_{alg} -near-optimal. Then

$$J(\hat{\alpha}) - J_* \leq 2\delta_J + \varepsilon_{\text{alg}}. \quad (\text{D.19})$$

Indeed, $J(\hat{\alpha}) \leq \tilde{J}(\hat{\alpha}) + \delta_J \leq \min \tilde{J} + \varepsilon_{\text{alg}} + \delta_J \leq J_* + 2\delta_J + \varepsilon_{\text{alg}}$. If the strict gap in (5.17) satisfies $g > 2\delta_J + \varepsilon_{\text{alg}}$, the same unique finite-set minimiser must be selected. This selection stability follows from the finite-objective gap and applies to the fixed candidate sets in Section 7 and Appendix E.

D.5. Legal restriction to another maturity

A continuous field certified on $[0, T_0]$ may be restricted to $[0, T]$, $T \leq T_0$, only while retaining the initial value and full history. The derivative-based reference exponent, curve/resolvent weights, reference centre, exact-transform modulus, strip bound and infinite-tail constants must be recomputed for the new maturity. Copying the half-year price centre or simply restarting the fractional equation is invalid. The quarter-year experiment uses this restriction and a separate complete pricing calculation; it does not assert generation of a new quarter-year residual field.

Appendix E. Complete experiments and verification duties

E.1. The separate historical three-candidate benchmark

These price and objective rows use the historical global-state bank, not the five-point nearby bank. They are preserved to explain the original financial task and failure boundary, with their original fixed quotes and outputs. Later matched propagation experiments change radii or reference configuration as specified in Section 7.

We use twelve quotes at the mathematical maturity $T = 1/2$ from the public SPX sample representing ordinary market conditions. Strikes range from 3700 to 4800 in steps of 100. The original CSV is identified by commit 860049da2b7486fe8aa509061eff23cc28c2ef89 in the WoonJeng data repository. The market bid and ask implied-volatility columns and the authors' model-output columns are identified separately. Market quotes define the objective; the error analysis in this paper supplies the model-price intervals.

The experiment uses normalised call prices with $D = 1, F = 4221.86$. Decimal IV and maturity values in the CSV are treated as exact rational inputs. Standard Black-Scholes conversion of the bid and ask IVs gives B_i, A_i . The target is the price midpoint $M_i = (B_i + A_i)/2$, rather than the Black-Scholes price obtained from the reported midpoint IV. These inputs jointly define the normalised quote setting below.

Fix $\rho = -.7445, \nu = .2897, \lambda_R = \kappa = 0$ and the entire forward variance function

$$\xi_*(t) = .0721 + (.0262 - .0721)E_{.5286}(-.5037t^{.5286}). \quad (\text{E.1})$$

The curve exponent .5286 and curve parameter .5037 remain fixed. Changing α neither recomputes the curve nor assigns .5037 to Riccati mean reversion. These values are the published rounded fitted parameters in [JK2021] and define a candidate comparison with the other inputs fixed.

The candidate set is

$$\Theta_{\text{finite}} = \{13/25, 3/5, 9/10\}, \quad H = \alpha - \frac{1}{2} \in \{.02, .1, .4\}. \quad (\text{E.2})$$

This is a conditional objective profile with all other parameters and the complete curve held fixed. The threshold $H_c = .1$ distinguishes the lower- and higher- H candidates in this comparison. All three satisfy $H < .5$, so the classification is within the roughness candidates rather than between rough and classical Heston models. All assertions in this section concern the complete three-element set (E.2).

The model prices $c_i(\alpha) = C_i(\alpha)/(DF)$ arise from the admissible nonnegative Volterra variance model and its exact affine transform. The model and kernel assumptions of AbiJaberElEuch2018v1 are verified individually in this paper; admissibility is established at the probability-model level. Define

$$J(\alpha) = \frac{1}{24} \sum_{i=1}^{12} [c_i(\alpha) - M_i]^2, \quad J_{\text{band}}(\alpha) = \frac{1}{24} \sum_{i=1}^{12} \text{dist}(c_i(\alpha), [B_i, A_i])^2. \quad (\text{E.3})$$

Thus the normalised price RMS is $\sqrt{2J}$, rather than $\sqrt{J/6}$. The same normalisation applies to the objective intervals, finite-candidate gap, and stability constants.

α	H	Rigorous model J interval	Rigorous model-price RMS interval	Rigorous implemented Padé J interval
0.52	0.0 2	[0.0000000068962, 0.0000001833669]	[0.00011744, 0.00060559]	[0.0000000708602, 0.000000708603]
0.60	0.1 0	[0.0000004535720, 0.0000006021720]	[0.00095244, 0.00109743]	[0.0000006127366, 0.0000006127367]
0.90	0.4 0	[0.0000082343900, 0.0000083484458]	[0.00405817, 0.00408619]	[0.0000081393603, 0.0000081393604]

All displayed endpoints are decimal outward roundings of rational bounds; the decisions use the untruncated exact rational endpoints. The unique minimiser of the model objective is $\alpha=0.52$, $H=0.02$. A strict lower bound on the gap between its objective and those of its competitors is 0.0000002702052. The implemented Padé output has the same unique finite-set minimiser, as certified by strict interval separation.

For $\varepsilon=0$, the outer set of exact optimal candidates is $\{13/25\}$. The specified higher-H candidates with $\alpha \geq 0.6$ are strictly separated from this optimum.

α	Rigorous model J'band interval	Rows strictly excluding quote consistency	All-row quote consistency certified
0.52	[0.0000000005732, 0.0000000877280]	[8, 9]	No
0.60	[0.0000003444775, 0.0000004648236]	[5, 6, 7, 8, 9, 10, 11]	No
0.90	[0.0000076345237, 0.0000077422463]	[1, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12]	No

Least-squares ordering and quote consistency are determined separately from objective intervals and rowwise price intervals. The following tables report the prices supporting these two model-validation results.

$\alpha=0.52$, $H=0.02$

K	Rigorous quote-midpoint interval	Rigorous normalised model-price interval	Implemented Padé dyadic output (displayed value)	Total model-price error bound for this output
3700	[0.13916863, 0.13916864]	[0.13882872, 0.13937153]	0.13910887	0.00028016
3800	[0.11921230, 0.11921231]	[0.11858902, 0.11913912]	0.11888921	0.00030019
3900	[0.09981335, 0.09981336]	[0.09909480, 0.09965209]	0.09943278	0.00033799
4000	[0.08123198, 0.08123199]	[0.08055552, 0.08111992]	0.08092667	0.00037115

K	Rigorous quote-midpoint interval	Rigorous normalised model-price interval	Implemented Padé dyadic output (displayed value)	Total model-price error bound for this output
4100	[0.06371452, 0.06371453]	[0.06322696, 0.06379836]	0.06361496	0.00038801
4200	[0.04759685, 0.04759686]	[0.04740368, 0.04798201]	0.04781654	0.00041287
4300	[0.03333802, 0.03333803]	[0.03349166, 0.03407684]	0.03393723	0.00044558
4400	[0.02173166, 0.02173167]	[0.02200137, 0.02259332]	0.02244498	0.00044361
4500	[0.01316937, 0.01316938]	[0.01332823, 0.01392686]	0.01373568	0.00040746
4600	[0.00760335, 0.00760336]	[0.00746696, 0.00807221]	0.00785225	0.00038529
4700	[0.00434644, 0.00434645]	[0.00393703, 0.00454881]	0.00430503	0.00036801
4800	[0.00253419, 0.00253420]	[0.00199815, 0.00261641]	0.00232396	0.00032582

$\alpha=0.60$, $H=0.10$

K	Rigorous quote-midpoint interval	Rigorous normalised model-price interval	Implemented Padé dyadic output (displayed value)	Total model-price error bound for this output
3700	[0.13916863, 0.13916864]	[0.13908096, 0.13925890]	0.13914793	0.00011096
3800	[0.11921230, 0.11921231]	[0.11900101, 0.11918134]	0.11908020	0.00010114
3900	[0.09981335, 0.09981336]	[0.09970355, 0.09988623]	0.09981101	0.00010746
4000	[0.08123198, 0.08123199]	[0.08138513, 0.08157015]	0.08152385	0.00013873
4100	[0.06371452, 0.06371453]	[0.06429081, 0.06447812]	0.06445106	0.00016025
4200	[0.04759685, 0.04759686]	[0.04869874, 0.04888832]	0.04888577	0.00018704
4300	[0.03333802, 0.03333803]	[0.03496352, 0.03515534]	0.03518748	0.00022397
4400	[0.02173166, 0.02173167]	[0.02351737, 0.02371142]	0.02375739	0.00024002
4500	[0.01316937, 0.01316938]	[0.01471126, 0.01490749]	0.01493438	0.00022313

K	Rigorous quote-midpoint interval	Rigorous normalised model-price interval	Implemented Padé dyadic output (displayed value)	Total model-price error bound for this output
4600	[0.00760335, 0.00760336]	[0.00857862, 0.00877702]	0.00878309	0.00020447
4700	[0.00434644, 0.00434645]	[0.00474038, 0.00494092]	0.00492883	0.00018845
4800	[0.00253419, 0.00253420]	[0.00255071, 0.00275338]	0.00270300	0.00015230

$\alpha=0.90$, $H=0.40$

K	Rigorous quote-midpoint interval	Rigorous normalised model-price interval	Implemented Padé dyadic output (displayed value)	Total model-price error bound for this output
3700	[0.13916863, 0.13916864]	[0.13880016, 0.13883098]	0.13876501	0.00006596
3800	[0.11921230, 0.11921231]	[0.11929811, 0.11932934]	0.11923074	0.00009859
3900	[0.09981335, 0.09981336]	[0.10073709, 0.10076873]	0.10063802	0.00013071
4000	[0.08123198, 0.08123199]	[0.08327826, 0.08331031]	0.08315433	0.00015597
4100	[0.06371452, 0.06371453]	[0.06710772, 0.06714016]	0.06696694	0.00017321
4200	[0.04759685, 0.04759686]	[0.05242401, 0.05245684]	0.05228161	0.00017523
4300	[0.03333802, 0.03333803]	[0.03943228, 0.03946550]	0.03931664	0.00014886
4400	[0.02173166, 0.02173167]	[0.02834450, 0.02837811]	0.02828552	0.00009258
4500	[0.01316937, 0.01316938]	[0.01934300, 0.01937699]	0.01935637	0.00002061
4600	[0.00760335, 0.00760336]	[0.01249183, 0.01252619]	0.01257890	0.00008707
4700	[0.00434644, 0.00434645]	[0.00765389, 0.00768862]	0.00780424	0.00015035
4800	[0.00253419, 0.00253420]	[0.00449079, 0.00452590]	0.00467714	0.00018635

The implemented Padé procedure is the selected six-condition GR [3/3] approximation. It uses a 256-point Jacobi rule for the derivative-based exponent and an eighth-order composite Gauss-Legendre rule for Fourier inversion with a fixed truncation limit of 200. The recorded files identify the Python/NumPy implementation

and exact source versions. Each finite numerical output c^P is saved as an exact dyadic rational. The model-price reference interval gives

$$|c^P - c| \leq \max\{|c^P - c^-|, |c^P - c^+|\}. \quad (\text{E.4})$$

This bound includes the implementation’s state, exponent, quadrature, truncation, and floating-point errors. The saved outputs can therefore be compared directly without a separate certification of an infinite-integral analytic Padé price. The numerical objective uses the same quote-price midpoint intervals. We assert selection preservation only when both the numerical and model objectives are strictly separated on the finite set and have the same minimiser. The assertion concerns the specified implementation, candidates, and input setting.

The historical global-state benchmark in this appendix uses the uniform state barrier proved in Appendix C.1. The finite-history alternatives and the new resolvent computation are stated in Section 3 and compared under identified configurations in Section 7.

E.2. Numerical identities, scope and evidence map

In the verification matrix only, the ID column labels proof obligations, rather than the configuration IDs of Section 2; obligation N2 covers the N2L local-bin refinement.

All configuration IDs and mathematical dimensions are fixed in configuration-registry.json. The following table states the actual reader obligation. Shared elementary interval primitives, shared continuous-derivative generator, reused fixed scientific input and an independently written assembly are distinct types of dependence. These are author-side acceptance checks; no outside referee execution is presumed.

I D	Obligation	Verification command	Recomputed	Inherited / shared
S 1	Padé coefficient signs on the original parameter rectangle	python bc-merged-20261007/full'structure'verify.py	211241 structural sign leaves and rectangle cover	model/parameter definitions; original strict interval algebra and generalized-power primitives
S 2	Startup Caputo residual and left-halfplane enclosure	python bc-merged-20261007/startup'independent.py	all 513 frequencies for each of the three original startup cells	stored field dyadics and later-time continuous residuals; Gamma/power/dyadic outward primitives
R 1	Full original alpha=.52 low-frequency continuous residual cover	python heston-nine-point-20261007/check-time-env elope.py	8189 closed cells x513 entries: cover, saved dyadic values and per-node maxima; startup audited separately	derivative inequalities in identified continuous generator; NumPy enclosure with proved dot-product error and strict scalar primitives

I D	Obligation	Verification command	Recomputed	Inherited / shared
R 2	Full $\alpha=.52$ high-frequency continuous residual cover	python heston-frontier-20 261007/omission-verify.py	8189 closed cells x512 entries: exact bank, startup/halfplane, saved-node maxima	identified Caputo derivative-generator inequalities; strict scalar primitives and original continuous generator
P 1	Global finite-history and low-frequency128-bin price accounts	python heston-nine-point- 20261007/verify-finite-his- tory-independent.py; python heston-nine-point- 20261007/independent-ti- me-local.py	forward masses, complete cell intersections, 513 tightened radii, all 1025 finite terms and complete spread remainder	continuous residual theorem/validity and identified stored field; strict forward-moment, exponent and true-tail primitives
P 2	Expanded-reference frozen half-year output account	python heston-frontier-20 261007/independent-omis- sion.py	changed reference centre, all 1025 support terms, old/new complete interval intersection, global/local high-frequency accounts	same actual fast output and continuous residual banks; original strict moment/scalar primitives
Q 1	Quarter-maturity used64 certificate and actual fast output	python heston-frontier-20 261007/independent-trans- fer.py --full	all three exact terminal interpolations, 1539 restricted exponents, three original fast-output calculations, all 3075 true finite CF/tail/price terms	larger-horizon continuous residuals and halfplane certificate; strict moment /trig/dyadic/true-tail libraries; original fast implementation used only for output replay
Q 2	Quarter full128 global and128-bin complete certificate	python heston-frontier-20 261007/independent-trans- fer-supplement.py --full	all 1025 restricted exponents, all 1025 CF/tail/coefficient/radius terms, all high bank entries, all 128 local weights and exact account	low/high continuous derivative validity; actual fast replay is separately performed by Q1; strict moment/trig/dyadic/true- tail libraries; path helpers of Q1
N 1	Fresh fields/residuals at the five nearby alpha points	python nearby`independent.py --N 1024; python nearby`independent.py --N 2048	all cells, startup, halfplane, 513 exponents and 1025 CF/coefficient terms per point; all objective/pair decisions	strict proof of continuous residual generator and final propagation theorem; declared sdk outward primitives, stored reference data
N 2	Descriptive128-bin reused-bank control, outside nearby objective grid	python local`output`indep- endent.py	128 exact positive weights, every intersecting closed-bank maximum, 513 tightened radii, all 12 prices and all actual output rounding	complete identity-matched N2048 point proof and continuous residual generator; same strict moment/trig/dyadic primitives

I D	Obligation	Verification command	Recomputed	Inherited / shared
C 1	Same model and complete tolerance for frozen/direct /corrected/BL-core outputs	python workload'controls' independent.py --N 2048 --local128	exact stored-output centre translation, allsame-reference remainder and28 portfolio supports; BL nominal-output metadata identity	reference certificate, BL nominal-output producer and its stored dyadics; same strict outward coefficient primitives

Exact generation commands, their empty-tree versus retained-input behavior, source hashes and boundary notes are in proof-obligation-matrix.json. Run commands in a disposable relative work copy to preserve the fixed packet. In particular, nearby'generate.py regenerates missing fields/residuals, but retains identity-matched components when present. nearby'independent.py can use an identity-matched per-point cache; the top-level fresh acceptance driver deletes every such cache before invocation.

For the explicitly retained V2 entrypoint source snapshots, the complete --full distinction is:

Entry point	Additional executed obligation	Outside that command
reproduce.py default	fixed-copy saved evidence readers; original frontier always requests bothquarter readers --full; new nearby reader point caches deleted by this top driver	all continuous residual generators or all structural signs
reproduce.py --full	default plus all211241 original structural sign leaves	fresh low/high/new-nearby continuous derivative generation
baseline/.../run'evidence.py --regenerate-continuous	original alpha=.52 513-node low continuous generator and retained8189-cell bank	new five-alpha reference/residual generation
baseline/.../run'frontier.py --regenerate-full	separate-clone full 512 high continuous generator plus fresh-bank audit	alter fixed quarter input receipts
nearby'generate.py --N N	missing field/residual/exponent components; otherwise reuses identified existing components	force regeneration over already retained scientific output
publication-v2 CI source.yml	source manifest identities and retained acceptance receipt binding	NumPy bank replay, interval arithmetic replay or residual generation

Thus the retained V2 reproduce.py --full adds structural-sign regeneration. It does not regenerate all continuous residual derivatives. Its source SHA is not imposed on the new V3 top-level driver. The original low continuous generator requires run'evidence.py --regenerate-continuous; the high-frequency generator requires run'frontier.py --regenerate-full, which uses a distinct reconstruction clone and preserves the fixed transfer input receipts. The historical CI checks source identities and the retained acceptance receipt binding; it does not perform the large-bank interval replay. A stored

PASS receipt, byte identities, executed recomputation and a mathematical theorem remain separately identifiable evidence.

The exact closest-pair gap equals the sum of the following signed contributions; scientific notation is a display only:

Gap contribution	Normalized loss contribution
reference`midpoint`objective`difference	7.939186982887e-09
used`node`support`both`candidates	-4.690368762320e-09
finite`zero`support`both`candidates	-1.217354923855e-09
strip`both`candidates	-1.855578458131e-12
true`infinite`tail`both`candidates	-9.568024410400e-11
reference`arithmetic`both`candidates	-6.928606526135e-23
midpoint`conversion`arithmetic`both`candidates	-4.770477111773e-27
candidate`a`quadratic`remainder	-1.195663126282e-09
box`intersection`endpoint`adjustment	0.000000000000e+00

The quarterly account has zero finite omissions; its infinite tail beyond 128 remains paid. The unused-node term in the nearby/global controls is nonzero and cannot be dropped by borrowing the quarter full-reference result. Exact ledger fractions and all 40 pair-mode records are delivered with independent secondary readback and intentional corruption controls.

Common ideal strict-reference floor in the BL-core complete bound (floors and percentages are approximate displays; exact fractions are authoritative):

Ban k	Nominal output	Reference floor points	Translation points	Floor / complete bound
N1	BL modified-Adams core, 512 steps	0.423484833	0.000853401	99.798887%
N1	BL modified-Adams core, 1024 steps	0.423484833	0.000305328	99.927953%
N2	BL modified-Adams core, 512 steps	0.261808768	0.000845646	99.678039%
N2	BL modified-Adams core, 1024 steps	0.261808768	0.000297572	99.886469%
N2L	BL modified-Adams core, 512 steps	0.228237113	0.000845646	99.630856%
N2L	BL modified-Adams core, 1024 steps	0.228237113	0.000297572	99.869791%

The separate returned-reference ratio also pays binary64 return rounding and is retained in the machine ledger. A small BL512/1024 nominal difference is an empirical comparison; its complete guarantee uses the same reference floor plus exact output translation. These floor fractions do not establish an intrinsic BL error estimate.

portfolio-pass-counts.svg and portfolio-pass-counts.pdf show the exact-endpoint pass-count step functions for the original three candidates and the regenerated alpha=.52 controls. Coordinates are converted to display decimals only after exact counting. The plotted 0-2-point range is stated explicitly; all thresholds are retained in

the CSV/JSON. Cross-bank curves are descriptive and within-bank methods share radii. Only typed scientific work counts, evidence bytes and version/hash identities are used; no host or duration measurements are part of this audit.

The matrix distinguishes the historical V2 driver snapshot from the current V3 entry. The V3 command additionally runs the separate dissipative-kernel reader, the exact experimental-account reader and bilingual extension-consistency checks. It still does not regenerate all upstream continuous derivative bounds.

New obligation	Generate	Verify	Separate recomputation and shared base
Dissipative weights and complete prices	v3/kernel/generate.py --baseline .	v3/kernel/independent.py --baseline .	14-layer direct series versus 12-layer moments; all 128 bins, 1025 nodes, 12 prices and 8 controls; strict primitives and upstream residual proof shared
Exact experiment ledgers and thresholds	v3/audit-experiments/prod uce'audit.py --baseline .	v3/audit-experiments/read' audit'independent.py --baseline .	Exact saved-bank accounts, pair gaps, 504 thresholds and 9 controls; inherited derivative validity
Integrated F/G mathematical text	Current editable appendix sources	v3/extensions/check'appen dices.py	Bilingual formula parity and scope checks; analytic proof review and retained classical readers are separately recorded

E.3. Budget sensitivity of every fixed direction

Figure E.1 uses exact endpoints for all 28 declared half-year directions. Solid curves denote joint certificates and dashed curves signed-marginal certificates; radii are matched within each configuration. Comparisons between different banks are descriptive. The shown range is zero to two index points; the machine table retains every threshold, including those beyond this plotting range.

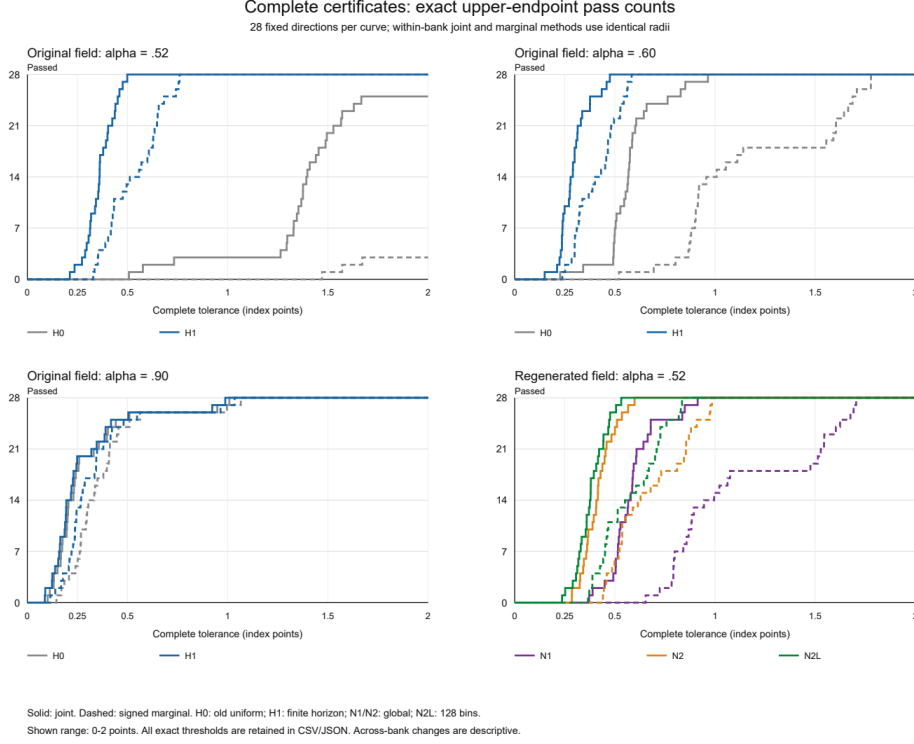


Figure E.1. Exact-threshold budget-pass-count curves; 28 fixed directions per configuration.

Appendix F. Original-chain classical Heston inclusion

This appendix supplies a second, classical-Heston upstream implementation of the joint-error interface. Its model-specific content is the passage from the correlated positive-part chain to a common original-law probability vector. The support-function and financial propagation steps use standard convex analysis. The terminal witness proves strict improvement of an outer set; it does not provide a complete annual monetary pricing certificate.

F.1. The two laws and admissible modes

Write $s = S/100 = e^Z$, $d = \kappa \bar{v}$, $r = 1/100$, $h = 1/768$, and $s_0 = 1$. The continuous law P and the original discrete law Q are

$$\begin{aligned}
 dZ_t &= (r - V_t/2)dt + \sqrt{V_t}(\rho dW_t + \sqrt{1 - \rho^2} dB_t), \\
 dV_t &= (d - \kappa V_t)dt + \xi \sqrt{V_t} dW_t, \\
 Y &= dh + (1 - \kappa h)v + \xi \sqrt{hv} G, \quad V' = Y^+, \\
 Z' &= z + rh - hv/2 + \sqrt{hv}(\rho G + \sqrt{1 - \rho^2} H).
 \end{aligned} \tag{F.1}$$

Here G, H are independent standard normal variables, and the same G drives both discrete updates. Throughout the common-moment argument,

$$\kappa \in [2, 4], \quad \bar{v}, v_0 \in [3/100, 3/50], \quad \xi \in [9/50, 7/25], \quad \rho \in [-4/5, -3/10]. \quad (\text{F.2})$$

A finite multirate mode has loadings $\alpha_{i,n} \in \mathbb{C}$ satisfying

$$\Re \alpha_{i,n} \leq 0, \quad \sum_n |\Re \alpha_{i,n}| \leq 1/2. \quad (\text{F.3})$$

The realized prefix is H_{ji} ; the remaining stock loading is $q_{ji} = p_{ji} + i\omega_{ji}$. All observation dates lie on the grid. A fixing moves one loading from the remaining sum to the realized prefix, without changing the underlying mode.

F.2. Exact continuation, the full kernel, and integrability

Lemma (exact continuous continuation). The affine continuation $u = e^{qz+a+bv}$, initialized with terminal $a = b = 0$, is the continuous conditional expectation of the mode. Between fixings, in remaining-time coordinates,

$$b' = \frac{1}{2}\xi^2 b^2 + (\rho\xi q - \kappa)b + \frac{1}{2}(q^2 - q), \quad a' = rq + db. \quad (\text{F.4})$$

At fixings q changes and a, b continue without resetting. To prove existence, set $\kappa_p = \kappa - p\rho\xi$, $\gamma_p = (p^2 - p)/2$. On $\Re b = 1$, the real drift is

$$\frac{1}{2}\xi^2 - \kappa_p + \gamma_p - \frac{1}{2}\{(\xi\Im b + \rho\omega)^2 + (1 - \rho^2)\omega^2\} < 0, \quad \frac{d|b|}{dt} \leq (\xi^2/2 - \kappa_p)|b| + |q^2 - q|/2. \quad (\text{F.5})$$

Indeed $p \in [-1/2, 0]$, $\kappa_p \geq 1.888$, and $\gamma_p \leq 3/8$. The first-crossing argument gives $\Re b \leq 1$; the radial estimate excludes explosion on every finite segment. Also $\Re a \leq (6/25)T$.

The identification with an expectation requires more than formal Riccati algebra. Multiply the real loadings by $5/4$, and let $L_*(t), p_*(t)$ be the resulting realized and remaining contributions. Then

$$\mathcal{Y}_t = e^{L_*(t) + p_*(t)Z_t + 4V_t - (24/25)t}, \quad |\mathcal{M}_\tau|^{5/4} \leq e^{(5/4)(6/25)T + (24/25)T} \mathcal{Y}_\tau. \quad (\text{F.6})$$

Here $\mathcal{M}$ is the stopped affine local martingale associated with the mode. The process $\mathcal{Y}$ is a nonnegative local supermartingale: its constant drift is nonpositive, and its variance drift is at most $65/128 - 4(1.86) + 8(7/25)^2 < 0$. Loading changes cancel at each fixing. Compact stopping and (F.6) give a uniform $5/4$ -moment bound. Uniform integrability removes stopping, proves the conditional-expectation identity, and covers $v = 0$ and every finite initial state.

The actual one-step propagation retains the entire positive-part Gaussian kernel. For next-step coefficients a, b, q , integration over H gives

$$\frac{Q_h u(z, v)}{e^{qz}} = e^{a + qrh - qhv/2 + q^2(1-\rho^2)hv/2} \int_{\mathbb{R}} \varphi(g) e^{q\rho\sqrt{hv}g + b[dh + (1-\kappa h)v + \xi\sqrt{hv}g]^+} dg. \quad (\text{F.7})$$

For $v > 0$ the integral splits at $g_0 = -(dh + (1 - \kappa h)v)/(\xi\sqrt{hv})$: the first part has variance exponent zero, and the second uses the positive candidate. Both parts are retained. At $v = 0$, $V' = dh$ deterministically. Completing a real Gaussian square is an integration device, not a replacement of Q by a mode-dependent probability law.

Let $D_{ji}(v)$ be this actual propagation minus the exact continuous continuation, after removal of the common $e^{q_{ji}z}$. Finite conditional telescoping, including each fixing exactly once, yields

$$\Phi_{Q,i} - \Phi_{P,i} = \sum_{j=0}^{N-1} r_{ji}, \quad r_{ji} = E_Q[H_{ji}e^{q_{ji}Z_j}D_{ji}(V_j)]. \quad (\text{F.8})$$

The integrability in the next subsection justifies every discrete expectation in this identity.

F.3. Original-law moments and shared occupation probabilities

Lemma (discrete exponential moment). For nonpositive real multirate loadings of total absolute value at most one,

$$E_Q \exp \left\{ \sum_{n \leq j} \beta_n Z_n + 4V_j \right\} \leq K_j := e^{6/25 + (73/75)jh}. \quad (\text{F.9})$$

To prove it, complete the real stock square for $p \in [-1, 0]$. The candidate mean becomes $dh + \eta_p v$, where $\eta_p = 1 - \kappa_p h \geq 191/192$. Since $(-y)^+ \leq h(25e)^{-1}e^{-25y/h}$,

$$E(-Y_p)^+ \leq \frac{h}{25e} \exp \left\{ -25d + \frac{[-25\eta_p + (625/2)\xi^2]v}{h} \right\} \leq \frac{h}{25e^{5/2}} < h/300. \quad (\text{F.10})$$

The parameter bounds imply $d \geq 3/50$, the variance coefficient is at most $-71/192$, and $e^{5/2} > 12$. The negative part is exactly zero at $v = 0$. For $B \geq 0$, $e^{By^+} \leq e^{By} + B(-y)^+$. Defining $F_p(B) = \gamma_p h + \eta_p B + \xi^2 h B^2/2$, Gaussian integration therefore gives

$$\begin{aligned} E_Q[e^{pZ' + BV'} \mid z, v] &\leq e^{pz + prh} \{e^{Bdh + F_p(B)v} + Bh e^{\gamma_p h v} / 300\}, \\ E_Q[e^{pZ' + 4V'} \mid z, v] &\leq e^{pz + 4v} (e^{4dh} + 4h/300) \leq e^{pz + 4v + (73/75)h}. \end{aligned} \quad (\text{F.11})$$

Here $F_p(4) \leq 4$, $rp \leq 0$, and $4d \leq 24/25$. Successive conditioning and cancellation of loading changes prove (F.9), using $4v_0 \leq 6/25$.

For a finite partition I_r of $[0, \infty)$, retain the zero atom and the unbounded tail, and set

$$\pi_{jr} = Q(V_j \in I_r), \quad h_{jir} \geq \sup_{v \in I_r} |e^{-2v} D_{ji}(v)|^2, \quad |r_{ji}|^2 \leq K_j \sum_r h_{jir} \pi_{jr} =: \mathbf{a}_{ji} \cdot \pi_j. \quad (\text{F.12})$$

The last inequality is complex Cauchy-Schwarz applied to $H_{ji}e^{q_{ji}Z_j}e^{2V_j}$ and $e^{-2V_j}D_{ji}(V_j)$; its first squared moment is covered by the doubled loadings in (F.9). Thus every mode uses the same original-chain probability vector. If a trial defect depends on additional states, one must bound uniformly over those states or enlarge the partition. Continuous-law residuals do not inherit these Q -probabilities.

F.4. A valid terminal probability polytope

Fix $\theta_* = (\kappa, \bar{v}, \xi, \rho, v_0) = (3, 9/200, 23/100, -11/20, 9/200)$, $j = 767$, $d = 27/200$, and $\eta = 255/256$. Use

$$I_0 = \{0\}, \quad I_r = ((r-1)/100, r/100] \quad (1 \leq r \leq 100), \quad I_{101} = (1, \infty). \quad (\text{F.13})$$

For $v > 0$ the next-step zero mass is $\Phi(-(dh + \eta v)/(\xi\sqrt{hv}))$; at $v = 0$ it is zero. Projection excess satisfies $E[-dh - \eta v - \xi\sqrt{hv}G]^+ \leq h\xi^2 e^{-d\eta/\xi^2}/(\eta\sqrt{2\pi e}) < h/200$. This follows by writing $v = hz$, bounding the normal negative part by its density term, and maximizing $\sqrt{z}e^{-\eta^2 z/(2\xi^2)}$. The mean recursion and the positive-exponential recursion give

$$EV_n \leq \mu := 7/150, \quad M_{n+1} \leq e^{4dh} M_n^\beta + h/50, \quad Ee^{4V_n} \leq 5/4, \quad Ee^{-tV_n} \geq e^{-t\mu}. \quad (\text{F.14})$$

Here $M_n = Ee^{4V_n}$, $\lambda = \kappa - 2\xi^2 = 14471/5000$, and $\beta = 1 - \lambda h \in (0, 1)$. Concavity of x^β gives the displayed recursion. At $M = 5/4$, use $\log(5/4) \geq 1/5$, $4d - \lambda/5 = -971/25000$, and $1 - e^{-x} \geq x/2$ for $0 \leq x \leq 1$; the decrease exceeds $h/50$. Since $M_0 = e^{-18} < 5/4$, induction proves the positive-exponential bound. Jensen gives the Laplace lower bound.

For $t \in \mathcal{T} = \{1, 4, 16, 64, 256, (191/192)^2/[2(49/625)h]\}$, define

$$t_0 = t, \quad t_{n+1} = \eta_* t_n - c_* t_n^2, \quad L_{767}(t) = \exp\left[-d_* h \sum_{n=0}^{766} t_n - v_* t_{767}\right]. \quad (\text{F.15})$$

The reference quadruple is $(\eta_*, c_*, d_*, v_*) = (191/192, (49/625)h/2, 3/50, 3/100)$, and the point quadruple is $(255/256, \xi^2 h/2, 27/200, 9/200)$. The bound $e^{-tY^+} \leq e^{-tY}$ and Gaussian integration give $E[e^{-tV'} | v] \leq e^{-dht} e^{-(\eta t - \xi^2 ht^2/2)v}$. All 767 recursion steps satisfy $0 \leq t_n \leq \eta_*/(2c_*)$. Consequently the recursion is monotone on the certified enclosures; the conservative reference choices yield a valid upper bound throughout (F.2).

The true terminal probability vector satisfies

$$\begin{aligned} \sum_r \inf_{I_r} e^{-tv} \pi_r &\leq L_{767}^{\text{ref}}(t), & \sum_r \inf_{I_r} e^{-tv} \pi_r &\leq L_{767}^{\text{point}}(t), \\ \sum_r \sup_{I_r} e^{-tv} \pi_r &\geq e^{-t\mu}, & \sum_r \ell_r \pi_r &\leq \mu, & \sum_r e^{4\ell_r} \pi_r &\leq 5/4, \end{aligned} \quad (\text{F.16})$$

together with nonnegativity and mass one; $\ell_r = \inf I_r$. The zero Laplace coefficient is one, and the tail infimum/supremum are zero/ e^{-t} . The six triples, two moment rows, and two signed mass rows give 22 inequalities. Upper-bound rows use lower coefficient endpoints and upper right endpoints; lower-bound rows use upper coefficients and lower right endpoints before sign reversal. Thus the saved rational $\mathcal{P}_{767} = \{\pi \geq 0 : A\pi \leq b\}$ contains the actual probabilities and is a nonempty compact subset of the simplex.

F.5. Certified terminal profiles on the entire variance axis

At the final step the future variance exponent is zero, so its positive-part correction vanishes exactly. For $q = p + i\omega$, write $g = (q^2 - q)/2$, $L = \rho\xi q - \kappa$, $c = \xi^2/2$. Then

$$D_q(v) = e^{rqh+hg v} - e^{a(h)+B(h)v}, \quad B' = g + LB + cB^2, \quad B(0) = 0, \quad a(h) = rqh + d \int_0^h B(t)dt. \quad (\text{F.17})$$

At θ_* , the Asian representative is $(p, \omega) = (-1/48, 64)$, and the put representatives are $(-1/4, -8), (-1/4, -24), \dots, (-1/4, -120)$. On $\Re B = 0$ the real drift is at most $p(p-1)/2 - (279/800)\omega^2 < 0$; hence $\Re B \leq 0$ and $|B(t)| \leq |g|t$.

Let $P_3(t) = b_1 t + b_2 t^2 + b_3 t^3$, where $b_1 = g$, $b_2 = Lg/2$, $b_3 = (L^2 g + 2cg^2)/6$. The exact guard $\Re b_1 + \max(\Re b_2, 0)h + \max(\Re b_3, 0)h^2 < 0$ proves $\Re P_3 \leq 0$. Its residual is $-\sum_{k=3}^6 r_k t^k$, with $r_3 = Lb_3 + 2cb_1 b_2$, $r_4 = c(2b_1 b_3 + b_2^2)$, $r_5 = 2cb_2 b_3$, $r_6 = cb_3^2$. Since $\Re[L + c(B + P_3)] \leq -\kappa_p$, variation of constants gives

$$E_B = \sum_{k=3}^6 |r_k|_+ \frac{h^{k+1}}{k+1}, \quad E_A = d \sum_{k=3}^6 |r_k|_+ \frac{h^{k+2}}{(k+1)(k+2)}. \quad (\text{F.18})$$

The notation $|\cdot|_+$ denotes a certified upper modulus. Set

$$\begin{aligned} A_q &= \min\{-d \int_0^h P_3|_+ + E_A, d|g|_+ h^2/2\}, \\ B_q &= \min\{|hg - P_3(h)|_+ + E_B, (|L|_+ + \xi^2|g|_+ h)|g|_+ h^2/2\}, \\ m_q &= 2 - \max\{h\Re g, \min(0, \Re P_3(h) + E_B)\} > 0. \end{aligned} \quad (\text{F.19})$$

The exponential-difference integral formula, and the coarse modulus bound for the two exponentials, imply

$$|e^{-2v} D_q(v)| \leq e^{prh} (A_q + B_q v) e^{-m_q v}, \quad h_{qr} = \min\{[4e^{2prh-4\ell_r}]_+, [e^{2prh} S_{qr}^2]_+\}, \quad S_{qr} = \sup_{v \in I_r} (A_q + B_q v) e^{-m_q v}. \quad (\text{F.20})$$

The supremum uses finite endpoints of each band closure and the stationary point $1/m_q - A_q/B_q$ when present. For $B_q = 0$, use the monotone branch without division. The tail limit is zero, and the zero band is evaluated at zero. Exact outward arithmetic verifies both Riccati guards and all 918 entries (nine modes, 102 bands), including the tail. The Asian conjugate has the identical profile.

F.6. Joint inclusion and its exact strictness criterion

Suppose the same finite mode catalog gives the complete discounted-price decomposition

$$e_k = p_{h,k} - p_{c,k} = \Re \sum_{j,i} c_{ki} r_{ji} + R_k, \quad |R_k| \leq \varrho_k. \quad (\text{F.21})$$

The coefficients and the remainder bounds must include all required conversion, truncation, and arithmetic contributions. The terminal witness alone does not establish this full-price premise. Let $\mathcal{F}$ be nonempty, compact, convex, and contain the actual full occupation tuple $\Pi = (\pi_j)_j$. Define

$$\mathcal{E} = \left\{ \left(\Re \sum_{j,i} c_{ki} z_{ji} \right)_k : \Pi \in \mathcal{F}, |z_{ji}|^2 \leq \mathbf{a}_{ji} \cdot \pi_j \right\} + \prod_k [-\varrho_k, \varrho_k]. \quad (\text{F.22})$$

Proposition (joint price inclusion). This set is nonempty, compact, convex, centrally symmetric, and contains the actual e . Its support is

$$s_{\mathcal{E}}(w) = \max_{\Pi \in \mathcal{F}} \sum_{j,i} \left| \sum_k w_k c_{ki} \right| \sqrt{\mathbf{a}_{ji} \cdot \pi_j} + \sum_k |w_k| \varrho_k. \quad (\text{F.23})$$

Proof. The lifted constraints are convex because $|z|^2$ is convex and the right side is affine; they are closed and bounded over compact $\mathcal{F}$. Their linear image plus the remainder box has the stated geometric properties. The actual (Π, r) is feasible by (F.12). For fixed Π , the support of each complex disk is its radius times the modulus of the real-linear coefficient. Independent disk phases attain the sum; the common maximization then gives (F.23). With cross-time constraints retain $\max_{\mathcal{F}} \sum_j$, rather than $\sum_j \max$; separation is justified only for $\mathcal{F} = \prod_j \mathcal{P}_j$. ■

Put $F_k(\Pi) = \sum_{j,i} |c_{ki}| \sqrt{\mathbf{a}_{ji} \cdot \pi_j}$, $m_k = \max_{\mathcal{F}} F_k$, and $\mathcal{B} = \text{rect}(\mathcal{E}) = \prod_k [-m_k - \varrho_k, m_k + \varrho_k]$. Then

$$s_{\mathcal{B}}(w) - s_{\mathcal{E}}(w) = \min_{\Pi \in \mathcal{F}} \left\{ \sum_k |w_k| [m_k - F_k(\Pi)] + \sum_{j,i} \left[\sum_k |w_k c_{ki}| - \left| \sum_k w_k c_{ki} \right| \right] \sqrt{\mathbf{a}_{ji} \cdot \pi_j} \right\}. \quad (\text{F.24})$$

Every summand is nonnegative. The gap is zero precisely when one common Π_* maximizes every active complete pricing row and all nonzero $w_k c_{ki}$ in each positive-radius disk lie on a common nonnegative complex ray. No phase condition is required at zero radius; $w = 0$ has zero gap. Adding and subtracting the common row values proves (F.24), and compactness makes the minimum attained. Absence of such a witness therefore proves strict improvement over the smallest coordinate box of the same outer set. It proves neither boundary attainment by the actual bias nor an actual-error lower bound. After payoff intersections, apply the criterion anew to the resulting set.

F.7. A complete terminal-row separation witness

The instruments are European puts at maturities $1/4, 1/2, 1$ and strikes $(90, 100, 110)$, followed by the twelve-fixing arithmetic Asian call spread $(A - 95)^+ - (A - 110)^+$, $A = \sum_{m=1}^{12} S_{m/12}/12$. The eighth instrument is the annual at-the-money put; $w = \mathbf{e}_{10} - \mathbf{e}_8$ is a prespecified bias comparison, not an optimized hedge.

After conjugate reduction, the complete terminal rows are

$$F_A(\pi) = C_A \sqrt{K_{767}} \sqrt{a \cdot \pi}, \quad C_A > 0, \quad F_P(\pi) = \frac{1600e^{-.01}}{\pi_{\text{circ}}} \sqrt{K_{767}} \sum_{\omega=8,24,\dots,120} \frac{\sqrt{d_{\omega} \cdot \pi}}{\sqrt{(\omega^2 + 1/16)(\omega^2 + 25/16)}}. \quad (\text{F.25})$$

Here π_{circ} is the circle constant, while π is the probability vector. The strict inequality $C_A > 0$ is substantive: the simplex Beta coefficient is $\mathcal{M}_K(\zeta) = K(12K/100)^{\sum_m \zeta_m} \prod_m \Gamma(\zeta_m) / \Gamma(2 + \sum_m \zeta_m)$. The original catalog retains the difference between strikes 95 and 110. Its Gamma factors are finite and nonzero, and the strike factors have unequal magnitudes $95(11.4)^{1/4}$ and $110(13.2)^{1/4}$, so at least one coefficient is nonzero.

The exact Asian primal and dual satisfy $\pi^A \geq 0$, $A\pi^A \leq b$, $y \geq 0$, $A^T y \geq a$, and $a \cdot \pi^A = b \cdot y$. Let $S = \{1, 2, 6\}$ and let J be the three positive-dual rows (point $t = 256$ Laplace, mean, mass). The exact compact expression for the witness is

$$\pi_S^A = A_{J,S}^{-1} b_J, \quad \pi_{S^c}^A = 0, \quad \det A_{J,S} \neq 0, \quad (A^T y - a)_{S^c} > 0. \quad (\text{F.26})$$

Its nonzero coordinates are approximately $((.0276833515163, .0487291439380, .923587504546))$. Weak duality proves optimality. Complementary slackness forces any optimizer onto S and onto the three active equalities; nonsingularity proves uniqueness. The strictly increasing square root transfers uniqueness to F_A .

The same polytope contains an exact π^B supported on $(-5, 6'')$. Let $D_\omega = (\omega^2 + 1/16)(\omega^2 + 25/16)$ and $x_\omega = d_\omega \cdot \pi^A > 0$. For the auxiliary put row $\mathcal{B}(\pi) = \sum_\omega \sqrt{d_\omega \cdot \pi} / \sqrt{D_\omega}$,

$$\left. \frac{d}{dt} \mathcal{B}((1-t)\pi^A + t\pi^B) \right|_{t=0} = \sum_\omega \frac{d_\omega \cdot (\pi^B - \pi^A)}{2\sqrt{D_\omega x_\omega}} \in [L, U], \quad L > 3 \cdot 10^{-9}. \quad (\text{F.27})$$

The exact rational L, U are the `put_direction.derivative` endpoints in `terminal767-result.json`. This reference specifies them exactly without a large printed fraction. Each coefficient enclosure $[l_\omega, u_\omega]$ obeys $0 < l_\omega \leq u_\omega$ and $4l_\omega^2 D_\omega x_\omega \leq 1 \leq 4u_\omega^2 D_\omega x_\omega$; negative profile changes reverse endpoints before summation. Thus the signed full-row derivative is positive. The Asian maximizer is not a put maximizer, so

$$g_{767}^{\text{row}} = \max F_A + \max F_P - \max(F_A + F_P) > 0, \quad s_{\mathcal{B}_{767}}(w) - s_{\mathcal{E}_{767}}(w) \geq g_{767}^{\text{row}} > 0. \quad (\text{F.28})$$

The second inequality follows from the coefficient triangle inequality; possible phase cancellation can only further reduce the joint support. With valid timewise product sets and the same complete catalog, the accumulated un-intersected support gap is at least this positive terminal gap. A modified catalog, probability set, or payoff intersection requires renewed analysis.

The machine appendix paths are `baseline/english-heston-release/code/classical/terminal767-input.json`, `terminal767-result.json`, `verify_input_bounds.py`, and `verify_terminal.py`. The first reader regenerates 22 rows, twelve 767-step Laplace recursions, and all 918 profile entries; the second verifies the 407 exact witness checks. These establish the terminal claims and leave the complete-price premise (F.21), annual monetary endpoints, and any trading interpretation unproved by this example.

F.8. Standard financial propagation corollary

For compact valid sets $e = p_h - p_c \in \mathcal{E}$ and $n = p_h - \hat{p}_h \in \mathcal{N}$,

$$\begin{aligned} w^T p_h - s_{\mathcal{E}}(w) &\leq w^T p_c \leq w^T p_h + s_{\mathcal{E}}(-w), \\ w^T \hat{p}_h - s_{\mathcal{N}}(-w) - s_{\mathcal{E}}(w) &\leq w^T p_c \leq w^T \hat{p}_h + s_{\mathcal{N}}(w) + s_{\mathcal{E}}(-w). \end{aligned} \quad (\text{F.29})$$

If the actual (n, e) belongs to a common compact $\mathcal{K}$, replace the lower subtraction by $s_{\mathcal{K}}(-w, w)$ and the upper addition by $s_{\mathcal{K}}(w, -w)$. For a closed calibration acceptance set $\mathcal{Y}$,

$$\begin{aligned}\mathcal{T}_{\theta} &= \{(n, e) \in \mathcal{K}_{\theta} : \widehat{p}_{h, \text{cal}} + n_{\text{cal}} - e_{\text{cal}} \in \mathcal{Y}\}, \\ \mathcal{A}_{\theta} &= \{\widehat{p}_{h, 10} + n_{10} - e_{10} : (n, e) \in \mathcal{T}_{\theta}\}.\end{aligned}\tag{F.30}$$

The identities $p_c = p_h - e = \widehat{p}_h + n - e$ prove these statements. Nonempty $\mathcal{T}_{\theta}$ is compact and attains the target endpoints; emptiness excludes compatibility with the acceptance rule. Enlarging the common input set to a coordinate box enlarges or preserves the target set. These are standard set-propagation consequences, conditional on a valid center and complete error inputs.

Appendix G. Nonexact trial fields and the signed error identity

This appendix extends the continuous-discrete comparison to one deterministic trial field that need not solve the backward equation. It states sufficient conditions independently of the unknown exact solution, proves the four signed terms, and gives an explicit globally integrable nonexact field. Expectations below are undiscounted; multiply the identity and its bounds by the common discount factor when converting a payoff to a price.

G.1. State, traces, and admissibility

Use the original laws P, Q , parameters, and correlated positive-part update of Appendix F. Between monthly fixings the state is $x = (s, v, A_m, \ell_m)$, where A_m is the sum of the realized normalized stock values and ℓ_m the sum of their logarithms. The fixing map J_i adds the current $s, \log s$ to these histories. Between fixings,

$$\mathcal{L} = rs\partial_s + (d - \kappa v)\partial_v + \frac{1}{2}vs^2\partial_{ss} + \rho\xi vs\partial_{sv} + \frac{1}{2}\xi^2 v\partial_{vv}.\tag{G.1}$$

Choose one fixed nonnegative weight for the entire horizon, for example

$$\begin{aligned}W &= e^v \{s^{-1/2} + e^{-\ell_m/24} s^{-(12-m)/24}\}, \\ W^{\text{nat}} &= e^v \{\mathfrak{B}_m^{-1/2} + \mathfrak{G}_m^{-1/2}\}, \quad \mathfrak{B}_m = (A_m + (12 - m)s)/12, \quad \mathfrak{G}_m = e^{\ell_m/12} s^{(12-m)/12}.\end{aligned}\tag{G.2}$$

Both weights match exactly across the fixing map. Since the arithmetic mean is at least the geometric mean, $W^{\text{nat}} \leq 2e^v \mathfrak{G}_m^{-1/2}$. The history factors in both weights are multivariate nonpositive real loadings with total absolute value at most $1/2$.

Lemma (weight moments). At θ_* , over $0 \leq t \leq 1$, either weight satisfies

$$\sup_t E_P W(t, X_t) < 5/2, \quad \max_j E_Q W(t_j, X_j) < 5/2, \quad \sup_{\tau \leq 1} E_P W(\tau, X_{\tau})^2 < \infty.\tag{G.3}$$

The last supremum is over the compact-localization stopping times used below. To prove the first bound for a branch, use the generator on e^{L+pZ+V} , with $p \in [-1/2, 0]$. Its variance coefficient is $\gamma_p - \kappa_p + \xi^2/2 < 0$, and its constant drift is $rp + d \leq d$. Loading changes cancel at fixings. Nonnegative stopped supermartingales give branch expectation at most $e^{v_0+d} = e^{9/50} < 5/4$. For the discrete law, (F.10)-(F.11) with $B = 1$

and $F_p(1) \leq 1$ give a branch bound $e^{v_0+d+1/300} = e^{11/60} < 5/4$. Sum two branches, or use $W^{\text{nat}} \leq 2e^v \mathfrak{G}_m^{-1/2}$. The strict comparisons follow from $\log(5/4) \geq 1/5$.

For the squared bound, $(a+b)^2 \leq 2(a^2+b^2)$. Each squared branch has total stock loading at most one and variance loading two. Increasing the latter to four dominates it; the continuous exponential-supermartingale variance coefficient is bounded by $1 - 4(1.776) + 8(7/25)^2 < 0$, with constant drift at most $24/25$. Stopping therefore gives a uniform finite second moment. The same argument covers the natural weight. This is enough for uniform integrability of fields dominated by W ; no ordinary martingale assumption on the stopped limiting stochastic integral is needed.

Let R be the same terminal payoff, or the same exact finite conversion remainder, under both laws. The deterministic field $\tilde{u}$ is locally $C^{1,2}$ between fixings, admits a right-sided extension at $v=0$ suitable for Itô's formula, and has genuine one-sided traces uniformly on compact state sets. Require global bounds

$$\begin{aligned} |\tilde{u}| &\leq CW, & |\mathfrak{r}(t, x)| &\leq \eta_c(t)W(t, x), & \int_0^1 \eta_c(t)dt &< \infty, \\ |d_i(x)| &\leq \eta_i W(t_i-, x), & |\delta(x)| &\leq \eta_T W(1, x), & C, \eta_i, \eta_T &< \infty. \end{aligned} \quad (\text{G.4})$$

Domination includes all states, $v=0$, and the unbounded tail. Define

$$\begin{aligned} \mathcal{D}_j &= Q_j \tilde{u}_{j+1} - \tilde{u}_j, & \mathfrak{r} &= (\partial_t + \mathcal{L})\tilde{u}, \\ d_i(x) &= \tilde{u}(t_i-, x) - \tilde{u}(t_i+, J_i x), & \delta &= R - \tilde{u}_N. \end{aligned} \quad (\text{G.5})$$

The jump defect is **left trace minus right trace after the fixing**. The operator Q_j includes the original kernel and any fixing at the end of the step. Use the same post-fixing terminal convention in both laws, and count each update once. The defects $\mathcal{D}_j$ must be integrable under the original Q ; a valid full-horizon enclosure is $\sum_j E_Q \mathcal{D}_j \in [L_Q, U_Q]$. A sufficient alternative is $|\mathcal{D}_j| \leq h\eta_{Q,j}W$.

These conditions do not define admissibility by assuming the desired error bound. An exact future-value field belongs to this class only when the stated regularity and domination are verified. A finite representation also belongs only after its growth, residuals, traces, terminal value, and original-kernel interface have been verified.

G.2. The four-term signed identity

Theorem (one-field residual identity). Under (G.3)-(G.5) and the integrable original- Q defects,

$$\boxed{E_Q R - E_P R = \sum_j E_Q \mathcal{D}_j - E_P \int_0^1 \mathfrak{r}(t, X_t)dt + \sum_i E_P d_i + (E_Q - E_P)\delta.} \quad (\text{G.6})$$

Proof. Finite conditional telescoping under Q gives

$$E_Q \tilde{u}_N - \tilde{u}_0 = \sum_j E_Q \mathcal{D}_j. \quad (\text{G.7})$$

For P , localize in compact state domains, on closed subintervals away from fixings. Itô's formula then has a stochastic integral of expectation zero. The second-moment

bound in (G.3) and $|\tilde{u}| \leq CW$ make the stopped field values uniformly integrable, so they converge in L^1 when stopping is removed. Tonelli and (G.4) give $E_P \int |\mathfrak{r}| \leq \int \eta_c(t) E_P W(t, X_t) dt < \infty$, which removes stopping from the time integral by absolute integrable domination. As the subinterval endpoints approach a fixing, compact-uniform genuine traces and continuous paths give almost-sure trace convergence; the same uniform integrability gives L^1 convergence. The actual field jump is $-d_i$. Summing all intervals and jumps therefore gives

$$E_P \tilde{u}_N - \tilde{u}_0 = E_P \int_0^1 \mathfrak{r}(t, X_t) dt - \sum_i E_P d_i. \quad (\text{G.8})$$

Subtract (G.8) from (G.7), and add the two terminal defects $\delta = R - \tilde{u}_N$. The common deterministic initial value cancels, giving (G.6). The argument also includes a fixing at the terminal date using the specified post-fixing convention. ■

If one instead defines $J_i^{\text{jump}} = \tilde{u}(t_i+, J_i x) - \tilde{u}(t_i-, x)$, its term in (G.6) is $-\sum_i E_P J_i^{\text{jump}}$. This explicit sign convention avoids mixing the two definitions. For an exact continuation, the backward residual, fixing defects, and terminal discrepancy vanish, recovering (F.8). For a nonexact field all four signed terms remain.

G.3. Effective sufficient bounds

At θ_* , (G.6) and the moment lemma imply

$$\begin{aligned} |E_Q R - E_P R| &\leq \max(|L_Q|, |U_Q|) + \frac{5}{2} \left(\int_0^1 \eta_c(t) dt + \sum_i \eta_i \right) + 5\eta_T, \\ |E_Q R - E_P R| &\leq \frac{5}{2} \left(h \sum_j \eta_{Q,j} + \int_0^1 \eta_c(t) dt + \sum_i \eta_i \right) + 5\eta_T \quad \text{if } |\mathcal{D}_j| \leq h\eta_{Q,j}W. \end{aligned} \quad (\text{G.9})$$

These are sufficient nonnegative bounds obtained by the triangle inequality. A joint signed enclosure of the four contributions can instead be propagated by its linear image. Separate price intervals alone supply no shared-state compatibility witness. In particular, a necessary lower bound on a residual envelope is not an evaluation of either sufficient upper bound in (G.9).

G.4. A self-contained nonexact analytic field

Fix $T = 1$, $\varepsilon > 0$, and $R = s_T^{-1/2}$. Consider

$$a_\varepsilon(t) = 1 + \varepsilon t(1 - t), \quad \tilde{u}_\varepsilon(t, x) = a_\varepsilon(t) s^{-1/2}, \quad 1 \leq a_\varepsilon \leq C_\varepsilon := 1 + \varepsilon/4. \quad (\text{G.10})$$

Use $W_A = s^{-1/2} e^v$, which is a single branch of the preceding weight proof. The field is globally regular for $s > 0, v \geq 0$, is independent of the histories, has exact matching fixing traces and terminal value, and obeys $|\tilde{u}_\varepsilon| \leq C_\varepsilon W_A$. Its generator residual is

$$\mathfrak{r}_\varepsilon = s^{-1/2} \{ \varepsilon(1 - 2t) + a_\varepsilon(t)(3v/8 - r/2) \}. \quad (\text{G.11})$$

Because $ve^{-v} \leq 1/e$, a global dominating function, constant in time, is

$$\eta_c = \varepsilon + C_\varepsilon(r/2 + 3/(8e)), \quad \int_0^1 \eta_c(t)dt = \eta_c < \infty, \quad \eta_i = \eta_T = 0. \quad (\text{G.12})$$

The original stock Gaussian integral is exact even though the stock and variance share G : this field does not depend on next-step variance. It gives

$$\mathcal{D}_j = s^{-1/2}\{a_\varepsilon(t_{j+1})e^{h(3v/8-r/2)} - a_\varepsilon(t_j)\}, \quad |\mathcal{D}_j| \leq h\eta_Q W_A, \quad \eta_Q = \varepsilon + C_\varepsilon\{r/2 + 3/[8e(1 - 3h/8)]\}. \quad (\text{G.13})$$

For the bound, write the difference as $(a_{j+1} - a_j)e^{h(3v/8-r/2)} + a_j(e^{h(3v/8-r/2)} - 1)$. Use $|a_{j+1} - a_j| \leq \varepsilon h$, $|e^x - 1| \leq |x|e^{\max(x,0)}$, and $ve^{-cv} \leq 1/(ec)$ for $c = 1 - 3h/8 > 0$. Thus every discrete defect is integrable, with the explicit full-horizon enclosure $\sum_j E_Q \mathcal{D}_j \in [-(5/2)\eta_Q, (5/2)\eta_Q]$, because $Nh = 1$. Substitution into (G.9) gives the finite sufficient bound $(5/2)(\eta_Q + \eta_c)$.

At $t = 1/2, v = r > 0$, (G.11) equals $-rC_\varepsilon s^{-1/2}/8 < 0$. Therefore this field is not the exact future-value function. The exact field for the same negative-power payoff exists by the affine continuation and moment proof of Appendix F, with loading $-1/2$. The same admissible object class contains both exact and nonexact fields; the four-term identity applies to each. This example is an analytic scope witness, not a numerical certificate for the original market instruments.

G.5. What the thirteen-function diagnostic does and does not certify

The saved classical trial-field diagnostic has a distinct, conditional role. With $n = 12 - m$, $B = (A_m + ns)/12$, $y = A_m/(A_m + ns)$, $\beta = 1 - y$, and $z = v/(1 + v)$, its branch is $B^{-q}P(y, v)$. In this subsection $\Re q = 1/2$, so B^{-q} corresponds to negative stock damping; this q is not the negative loading q used in Appendix F. The conjugated generator is

$$\begin{aligned} \mathcal{L}_q P = & -ry\beta P_y + dP_v - qr\beta P \\ & + v\{\tfrac{1}{2}y^2\beta^2 P_{yy} - \rho\xi y\beta P_{yv} + \tfrac{1}{2}\xi^2 P_{vv} + (1+q)y\beta^2 P_y \\ & - (\kappa + q\rho\xi\beta)P_v + \tfrac{1}{2}q(q+1)\beta^2 P\}. \end{aligned} \quad (\text{G.14})$$

The basis consists of $\phi_{ij} = \beta y^i z^j$, $0 \leq i \leq 2, 0 \leq j \leq 3$, and $\psi = v\beta^2$. It exactly represents the terminal forcing $\mathcal{L}_q 1 = -qr\phi_{00} + q(q+1)\psi/2$. This is completion of the terminal generator, not closure under every generator image. For finite endpoint constants, the field uses cubic Hermite interpolation with exact physical slopes $-Kc - g(q)$, where $g(q) = -qre_{00} + q(q+1)e_\psi/2$. The exact fixing carry, with $\gamma = (n-1)/n$, is

$$(T_n c)_{\ell j} = \gamma \sum_{i=\ell}^2 \binom{i}{\ell} \gamma^\ell n^{-i+\ell} c_{ij}, \quad T_n \psi = \gamma^2 \psi. \quad (\text{G.15})$$

It follows from $y' = \gamma y + 1/n$, $B' = B$, and $\beta' = \gamma\beta$. Terminal correction nodes are zero. Exact carries and slopes specify the traces; floating generation values do not replace these definitions. The finite field has $|P| \leq C_0 + C_1 v \leq (C_0 + 2C_1/e)e^{v/2}$. Its continuous residual is a polynomial of degree at most two in v with bounded y, z coefficients, so $e^{-v}|F| \leq a_0 + a_1/e + 4a_2/e^2$. A complete application of (G.9) still

requires the actual coefficient bank, all-state original- Q envelopes, and full-horizon sufficient upper bounds.

For the saved first-month state $m = 0, y = 0, v = 9/200$, let $w = (1, z, z^2, z^3, 0, \dots, 0, v)$, ℓ be the generator-value row of the basis, and $\ell_0 = -qr + q(q+1)v/2$. Exact integration over one piece gives

$$\int F_{A,q} dt = \left[w + \frac{h^2}{12} \ell K \right] (c_R - c_L) + \frac{h}{2} \ell (c_L + c_R) + h \ell_0. \quad (\text{G.16})$$

Indeed $\int c(t) dt = h(c_L + c_R)/2 + h^2(s_L - s_R)/12$, and $s_L - s_R = K(c_R - c_L)$. The archive contains all 3080 weighted complex integral intervals (385 modes and eight cells, covering the original 64 time pieces). At common phase zero, sum the real mode intervals on each cell, apply $\int |F| \geq |\int F|$, and subtract the integrated geometric-branch upper bound. The exact aggregate is specified by the mathematical field `whole_lower` in `field-result.json`:

$$\int_0^{1/12} \eta_c(t) dt \geq L_*, \quad L_* > 0.003407444052031154 > 0. \quad (\text{G.17})$$

This rejects the selected allocation $\int \eta_c \leq 1/1000$, and retains the status `FAIL_SELECTED_INTEGRATED_PDE_CONTRACT_ONLY`. It does not reject every sufficient allocation, minimize over the trial space, or bound the actual signed price bias from below. The portable `verify_field_receipt.py` checks exact saved interval aggregation; it does not regenerate the absent coefficient bank or rerun the archived direct integration. Accordingly, (G.17) is an archived-field necessary-condition diagnostic, while the explicit field (G.10) independently proves nonexact admissibility without that bank. No complete annual monetary PASS is inferred.

The machine appendix uses `baseline/english-heston-release/code/classical/field-cell-integrals.json`, `field-independent-readback.json`, `field-result.json`, `provenance.json`, and `verify_field_receipt.py`, with publication manifests fixing their scientific identities. Contact, environment, and execution-measurement metadata are unnecessary to these mathematical assertions.

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